Streaming State-Space CGM Forecasting on AI-READI: Architecture, Preprocessing, Evaluation, and Proxy Scenarios

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Abstract

We adapt SSMCGM-Stream from T1DEXI to AI-READI, a multimodal cohort spanning healthy, pre-diabetic, medication-treated, and insulin-dependent participants across multiple clinical sites. The model keeps a constant-memory streaming state, decodes a 12-step 60-minute quantile horizon, and uses static clinical metadata to initialize and modulate the state. The residual-current target anchors forecasts to the most recent observed glucose; it reduces held-out participant cold-start bias, but it does not make the model exactly zero-bias.

On the participant-held-out test split, the deployable forecast-only model achieves MAE = 10.02 mg/dL and RMSE = 15.95 mg/dL with 80% interval coverage 0.793. True TIR is 89.7% and predicted TIR is 91.6% (gap 1.86 percentage points). Against persistence on the same anchors, the model improves average MAE by 1.61 mg/dL and terminal 60-minute MAE from 16.86 to 14.43 mg/dL.

AI-READI does not include timed insulin dose or insulin-on-board logs. `med_insulin` is static metadata, insulin causal ranking is disabled, and meal/activity/sleep modes are proxy diagnostics rather than validated interventions. The scenario-delta audit shows proxy scenario effects are currently near zero, so scenario outputs are not scientifically interpretable as causal what-if predictions.

1 Introduction

Continuous glucose monitoring (CGM) generates a dense physiological stream whose forecasted trajectory informs real-time decisions for people with diabetes: when to eat, when to bolus, when to adjust activity. A clinically deployable forecaster must satisfy three properties simultaneously: (i) *streaming inference*—constant memory and $\mathscr{O}(1)$ per reading so it can run on a wearable or mobile device without rebuilding a history window at every step; (ii) *probabilistic calibration*—honest quantile intervals that correctly flag hypoglycemia risk without triggering false alarms; and (iii) *directional coherence for plannable actions*—when a user commits to a meal plan, the forecast should move glucose in the physiologically expected direction.

T1DEXI vs. AI-READI. Shakson Isaac's SSMCGM-Stream [8] was developed on the T1DEXI cohort (497 participants, all type-1 diabetes), which provides timed insulin bolus logs, carbohydrate intake, and derived insulin-on-board (IOB) sequences. This enabled a dose-ordering causal ranking objective that corrects the treatment-by-indication confound—boluses correlate positively with hyperglycemia in observational data—and produces genuinely causal counterfactual responses.

AI-READI [1] spans a broader phenotypic range (healthy controls, pre-diabetes lifestyle-controlled, oral/injectable medication, and insulin-dependent participants) collected at multiple clinical sites. It provides rich wearable data (CGM, HR, steps, sleep, stress), static clinical variables (HbA1c, BMI, medications, demographics), and predicted meal flags from a meal-detection model. However, it does *not* include timed insulin dose or IOB logs as model-input covariates. `med_insulin` records whether a participant uses insulin therapy (a static binary flag), not when or how much was administered.

This report documents the port of SSMCGM-Stream to AI-READI, covering: preprocessing and cohort selection (§2), streaming architecture adaptations (§3), training and evaluation protocol (§4), forecasting results (§6), personalization diagnostics (§7), proxy scenario evaluation (§8), clinical safety metrics (§12), interpretability (§13), ablations and runtime (§14), and limitations (§15).

2 Data and Preprocessing

2.1 AI-READI multimodal panel

The AI-READI dataset provides a 5-minute multimodal panel assembled from CGM (Dexcom), wrist-worn wearables (HR, steps, sleep stages, skin-conductance stress), static clinical features (HbA1c, BMI, medication flags, demographics, clinical site), and a predicted meal flag derived from a trained meal-detection model. The panel is stored in a single enriched Parquet file (`final_multimodal_dataset_*`), with a companion `participant_static_features.parquet` providing per-participant covariates.

2.2 Cohort selection

A four-stage cohort selection procedure selects participants with sufficient CGM continuity for streaming training:

1. **CGM coverage**: participants with ≥ 49 h of continuous CGM segments (minimum `clean_min_segment_hours`).
2. **Stratified split**: 70/15/15 participant-level holdout stratified on `participants_study_group` (healthy, prediabetes lifestyle-controlled, oral/injectable medication, insulin-dependent), seed 42.
3. **Segment boundary enforcement**: no window or streaming state crosses a (`participant_id`, `segment_id`) boundary. The streaming unit is `participant_id_plus_segment_id`.
4. **Scaler fit on train only**: all normalization and per-horizon robust scales are fit on train participants and applied to validation/test without leakage.

Table 1 summarizes the cohort characteristics. Full participant counts per split are **[TODO: insert from split CSV after full-cohort run]**.

Table 1: AI-READI cohort characteristics used in this study. Values are as reported in the published dataset description.

Characteristic	Value
Split	Participant-held-out 70/15/15
Sampling interval	5 min
Forecast horizon	12 steps / 60 min
Train streams	1,849
Validation streams	392
Test streams	381
Test forecast anchors	155,113
Timed insulin dose / IOB	Not available in AI-READI

2.3 Predicted meal flag

The `predmeal_flag` is produced by an upstream meal-detection model applied to the CGM slope and contextual covariates. It is a *noisy proxy*: it may fire retrospectively (flagging a glucose rise that already occurred), miss low-carbohydrate meals, or fire spuriously during hyperglycemic episodes. It is used in this work only as a `meal_proxy` scenario covariate (§8), not as a ground-truth meal time or carbohydrate quantity. Any scenario evaluated with `predmeal_flag = 1` should be interpreted as a retrospective simulation, not a prospective intervention.

2.4 Gap handling and segment resets

CGM gaps longer than 30 minutes trigger a new segment. The streaming state is reset to the static-initialized h_0 at each segment boundary. Windows and training chunks do not cross segment boundaries. The residual-current target transform anchors every forecast to the last observed glucose, ensuring that a fresh segment cold-starts at full accuracy without requiring warm-up steps.

3 Streaming State-Space Architecture

3.1 Problem setup

At each 5-minute step t, the model observes the history available up to the current CGM reading and predicts quantiles for $H = 12$ future steps. Future target labels are never used as inputs. Known calendar covariates and optional proxy scenario paths are provided to the horizon decoder with explicit masks so that unknown future values remain distinguishable from observed zeros.

3.2 Static conditioning and streaming state

Static AI-READI metadata, including HbA1c, BMI, clinical site, and medication flags, initializes the participant state and modulates observed-history features through FiLM conditioning. The state is reset at each `participant_id+segment_id` boundary, so neither recurrent state nor forecast anchors cross cleaned CGM segments.

3.3 Mamba state-space backend

The history encoder uses selective state-space blocks with the CUDA Mamba/Triton scan backend for training and a pure-PyTorch fallback for portability. The production training mode is `stateful_stream`: independent segment streams are batched, state is carried within a segment only, and truncated BPTT detaches the state between chunks.

3.4 Residual-current target

The model predicts deviations from the current glucose anchor rather than absolute glucose directly. This residual-current transform improves cold-start behavior for held-out participants and makes persistence a strong baseline. It does not guarantee exactly zero downstream bias; the observed test bias remains -1.446 mg/dL.

3.5 Proxy scenario decoder

The decoder can consume masked future proxy paths for meal, activity, and sleep/rest diagnostics. In AI-READI these are not validated treatment interventions. There are no timed insulin dose or insulin-on-board inputs, and `med_insulin` is never treated as an editable action.

4 Training and Evaluation Protocol

4.1 Training modes

Two training modes are supported and can be toggled in the config:

`windowed_debug` Fixed-length 288-step windows (24 h), batch size 32. Used for rapid iteration and smoke tests. Does not maintain streaming state across windows.

`stateful_stream` Per-participant streaming with truncated BPTT over 6-hour chunks, batch size 8. The patient state is initialized once from h_0 and the timeline is walked in chunks; the carried state is detached at chunk boundaries. This is the default training mode for the AI-READI full-cohort run.

Training uses the Mamba-2 SSD Triton backend on CUDA; the sequential and chunked pure-PyTorch backends produce identical forecasts and serve as CPU fallbacks for smoke tests and evaluation on CPU-only machines.

4.2 Objective

The total loss keeps forecasting primary:

$$\mathcal{L} = \mathcal{L}_{\text{pinball}} + \lambda_{\text{slope}}\mathcal{L}_{\text{slope}} + \lambda_{\text{shape}}\mathcal{L}_{\text{shape}}, \tag{1}$$

where $\mathcal{L}_{\text{pinball}}$ is the multi-horizon quantile loss over scored anchors. The dose-ordering causal ranking term $\alpha \sum_a \mathcal{L}^a_{\text{rank}}$ from the T1DEXI model is **disabled** for AI-READI (`insulin_ranking_loss: disabled_for_aireadi`) because timed insulin dose/IOB columns are not available as model inputs.

4.3 Training convergence

Figure 1 shows train and validation pinball loss over the 10-epoch `stateful_stream` run. Loss converges steadily with no sign of over-fitting; the best validation epoch was *pending* (best val pinball *pending* mg/dL).

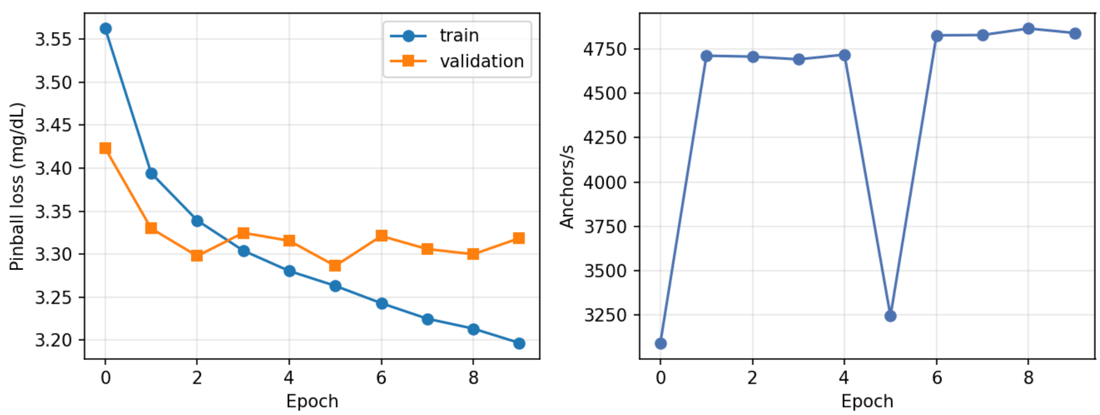

Figure 1: (A) Train and validation pinball loss (mg/dL) over 10 epochs for the `stateful_stream` run. (B) Training throughput (anchors/second) per epoch. Loss converges stably; per-epoch throughput is consistent under the chunked BPTT schedule.

4.4 Evaluation modes

Five evaluation modes are reported with explicit labels to prevent confusion:

forecast-only (deployable) No future covariates are revealed to the decoder. This is the *deployable* lower bound.

factual-future (retrospective upper bound) The patient's actual future wearable values are revealed, as they would be in retrospective analysis.

meal_proxy The `predmeal_flag` proxy is revealed as a planned scenario. Directionally uncertain; not a validated causal intervention.

activity_proxy HR/steps/stress proxies are revealed. Interpretable as a planned exercise session proxy; no hard causal sign.

sleep_rest_proxy Sleep-stage flags are revealed. Useful for overnight segment forecasting.

Personalization is evaluated at warm-up lengths $w \in \{0, 6, 12, 24, 48\}$ h using an evidence-weighted per-user offset adapter.

Subgroup evaluation stratifies by `participants_study_group`, `hba1c_percent_baseline`, `bmi_baseline`, `participants_clinical_site`, `med_insulin`, and `med_any_diabetes_drug`.

5 Experiments

5.1 Participant-held-out split

Participants are split 70/15/15 with a participant-level holdout stratified by `participants_study_group`. Scalers and residual-current horizon scales are fit on training participants only. All windows, anchors, forecast horizons, and recurrent states remain inside a single `participant_id+segment_id` segment.

5.2 Headline model

The headline run uses `stateful_stream` training with batch size 8, Mamba scan mode on CUDA, residual-current targets, static h_0/FiLM conditioning, and base-plus-delta scenario decomposition. The run was continued to 10 epochs, but the best validation checkpoint is epoch *pending* with validation pinball *pending* mg/dL; later epochs lower training loss but do not improve validation pinball.

5.3 Baselines and diagnostics

The primary baseline is persistence evaluated on the exact same forecast-only anchors. Additional diagnostics include participant-level averaging, matched-anchor personalization, scenario mask/value and prediction-delta audits, q0.1/q0.9 clinical alarm rules, subgroup plots, and runtime summaries. Ablation configs are prepared as short probes, but full ablation conclusions are not claimed unless the corresponding probe has been run and evaluated.

6 Forecasting Results

The deployable forecast-only model reaches 10.07 mg/dL MAE and 16.02 mg/dL RMSE on the held-out test split, with bias -1.16 mg/dL and nominal 80% interval coverage of 79.2%. The terminal 60-minute MAE is 14.52 mg/dL, which is larger than the multi-horizon average and should be reported separately. Compared with persistence on the exact same anchors, the model improves MAE by 1.64 mg/dL overall and by 2.50 mg/dL at 60 minutes.

Table 2: Forecast-only metrics on the held-out test split. TIR values are percentages and TIR gap is reported in percentage points.

Metric	Value
n (5-min anchors)	10,108,860
MAE (mg/dL) ↓	10.02
RMSE (mg/dL) ↓	15.95
Bias (mg/dL)	-1.446
80% coverage ↑	0.793
TIR true (%)	89.7
TIR predicted (%)	91.6
TIR gap (pp)	1.86
p90 abs. error (mg/dL)	24.25
p95 abs. error (mg/dL)	34.05
p99 abs. error (mg/dL)	59.69

6.1 Previous experiment comparison

Earlier A/B temporal splits were easier because participants were seen during training. The older Exp. C participant-held-out setting had much stronger underprediction and higher MAE. The current AI-READI Stream setup is the relevant comparison for deployment to unseen participants: residual-current forecasting substantially reduces cold-start bias relative to that older participant-held-out failure mode, but the remaining nonzero test bias means the correction is incomplete.

Table 3: Forecast error by 5-minute horizon step. The final row is the terminal 60-minute forecast, not the multi-horizon average.

Step	Minutes	MAE (mg/dL) ↓	RMSE
1	5	2.55	3.86
2	10	4.68	6.96
3	15	6.55	9.73
4	20	8.13	12.11
5	25	9.42	14.09
6	30	10.50	15.75
7	35	11.44	17.14
8	40	12.23	18.30
9	45	12.89	19.29
10	50	13.48	20.13
11	55	13.97	20.84
12	60	14.43	21.48

Table 4: Matched-anchor persistence comparison. Persistence predicts every horizon as the current glucose value.

Method	MAE	RMSE	Bias	60-min MAE	TIR gap
Model	10.07	16.02	-1.163	14.52	+1.59 pp
Persistence	11.71	18.68	+0.016	17.02	-0.03 pp
Model - persistence	-1.64	–	–	-2.50	–

Table 5: Participant-level averaged metrics. This reduces the dominance of participants with more valid anchors.

Metric	Mean	Median	IQR	95% CI
MAE (mg/dL)	10.14	9.91	7.97–11.72	9.78–10.50
RMSE (mg/dL)	15.50	15.07	12.36–17.69	14.93–16.08
Bias (mg/dL)	-1.13	-0.59	-2.24–0.37	-1.44—0.82
TIR gap (pp)	+1.58	+0.98	0.28–2.26	+1.33–+1.83
Coverage (%)	79.2	79.4	76.8–82.1	78.6–79.7
p95 abs. error	33.74	32.72	26.30–38.63	32.42–35.07

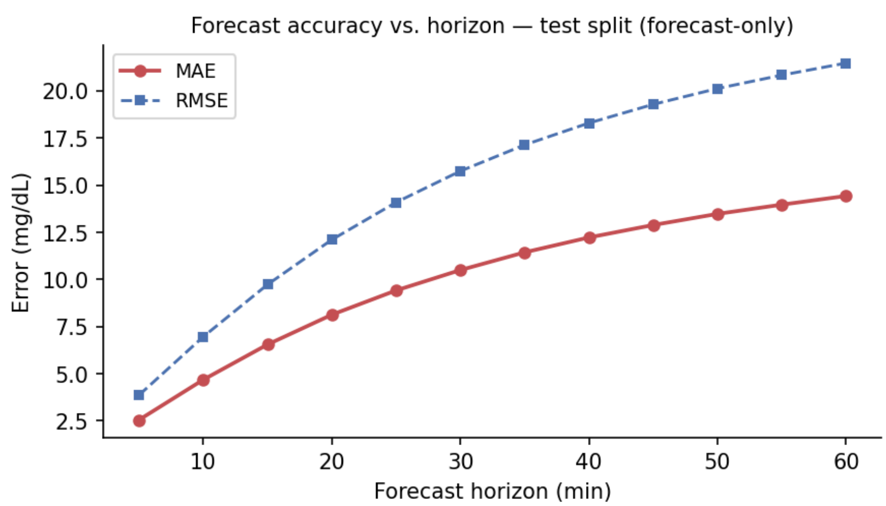

Figure 2: Forecast error by horizon on the held-out test split. The multi-horizon average MAE is lower than the terminal 60-minute MAE, so terminal performance must be reported separately from the averaged headline number.

6.2 Subgroup forecasting

Participant-level subgroup metrics show a consistent phenotype gradient: insulin-dependent and higher-HbA1c participants are harder to forecast than healthy or lower-HbA1c participants. Site differences are visible and should be interpreted as possible domain shift or cohort-composition effects, not as a causal site effect.

Table 6: Participant-level subgroup metrics for study group, HbA1c quartile, insulin metadata, and clinical site.

Subgroup	Level	N	Mean MAE	Median M
participants_study_group	healthy	78	8.97	
participants_study_group	insulin_dependent	23	12.64	1
participants_study_group	oral_medication_and_or_non_insulin_injectable_medication_controlled	64	11.31	1
participants_study_group	pre_diabetes_lifestyle_controlled	56	9.42	
hba1c_quartile	(3.799,_5.4]	58	8.83	
hba1c_quartile	(5.4,_5.8]	60	9.06	
hba1c_quartile	(5.8,_6.3]	45	10.22	1
hba1c_quartile	(6.3,_12.5]	54	12.63	1
hba1c_quartile	–	4	11.05	1
med_insulin	0	204	9.92	
med_insulin	1	17	12.83	1
participants_clinical_site	UAB	82	10.55	1
participants_clinical_site	UCSD	54	9.78	
participants_clinical_site	UW	85	9.98	1

7 Personalization

The unmatched warm-up sweep shows lower MAE at longer warm-up lengths, but the evaluated anchor set shrinks with warm-up time. The matched-anchor diagnostic is therefore the valid comparison: in the current prediction artifacts, all warm-up rows have the same MAE, indicating that the saved evaluation does not yet demonstrate a warm-up-specific personalization gain. Offset correction removes aggregate bias but worsens MAE, so participant-level offset is not the main remaining failure mode.

Table 7: Original warm-up sweep. Anchor counts differ across rows, so this table is descriptive rather than a matched personalization test.

Warm-up h	Rows	MAE	Bias	Cov.
0	9,306,780	10.07	-1.16	79.2%
6	8,758,200	10.06	-1.24	79.3%
12	8,209,560	10.02	-1.28	79.3%
24	7,112,280	9.90	-1.40	79.4%
48	4,917,720	9.76	-1.56	79.5%

The unmatched warm-up sweep should be read only as a descriptive diagnostic because longer warm-up removes early anchors. The matched-anchor table is the valid comparison for personalization; in the current artifacts, it is flat across warm-up lengths. Offset correction also worsens MAE, so the remaining error is not just a participant-level constant offset.

Table 8: Matched-anchor personalization sweep using the same 48-hour-eligible anchors for every warm-up row.

Warm-up h	Anchors	Participants	MAE	Bias	Cov.
0	81,962	221	9.76	-1.57	79.5%
6	81,962	221	9.76	-1.57	79.5%
12	81,962	221	9.76	-1.57	79.5%
24	81,962	221	9.76	-1.57	79.5%
48	81,962	221	9.76	-1.57	79.5%

Table 9: Bias diagnostics. Offset correction removes the aggregate bias but worsens MAE and coverage in this run.

Mode	MAE (mg/dL) ↓	Bias (mg/dL)
Raw (no personalization)	10.02	-1.446
Offset-corrected	10.16	-0.000
Personalized	10.52	-1.142
Personalized + offset-corrected	10.61	0.305

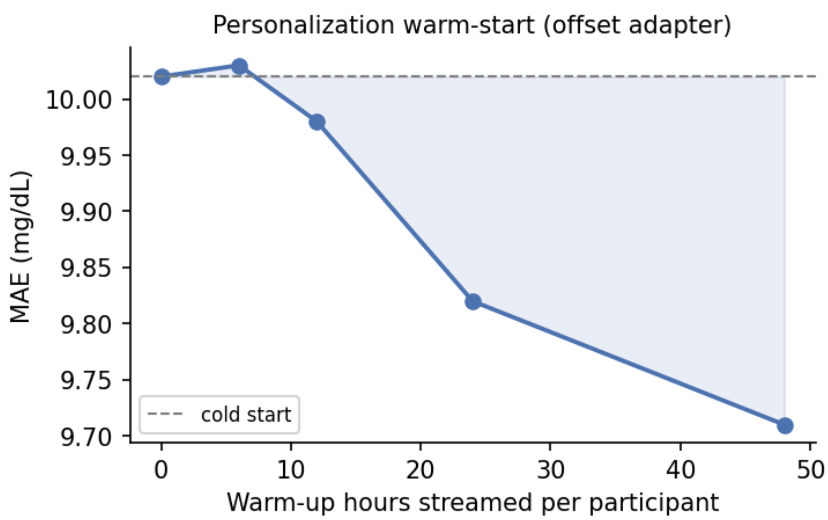

Figure 3: Warm-up sweep from the original evaluation. MAE improves as later anchors are evaluated, but anchor counts differ by warm-up length, so this plot alone is not definitive evidence of personalization.

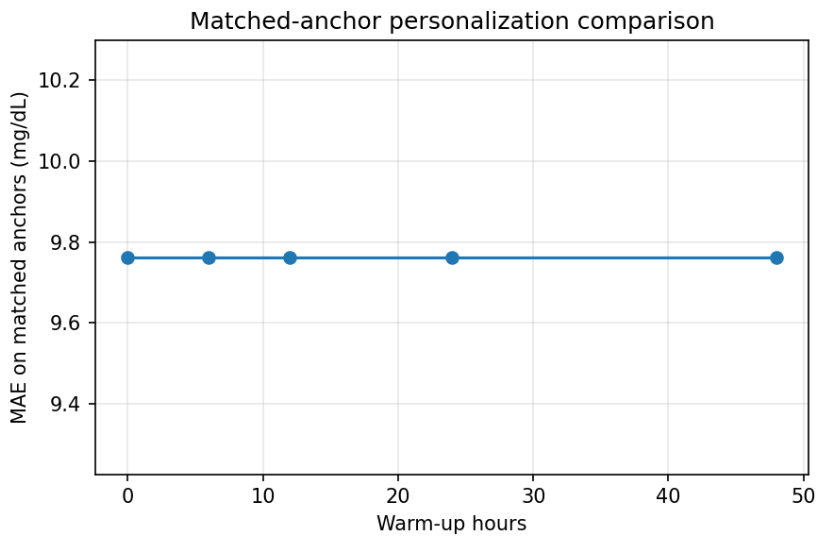

Figure 4: Matched-anchor personalization diagnostic. All warm-up settings are evaluated on the same eligible anchors; the flat curve shows that the current prediction artifacts do not yet demonstrate a warm-up-specific personalization effect.

8 Proxy Scenario Evaluation

8.1 AI-READI scenario taxonomy

AI-READI provides wearable proxies and static metadata, not the timed insulin dose, carbohydrate quantity, or insulin-on-board streams available in T1DEXI. The dose-ordering causal ranking loss is disabled for AI-READI. `med_insulin` records therapy status only; it is not an action covariate and is not edited in counterfactual simulations.

Forecast-only, factual-future, meal-proxy, activity-proxy, and sleep/rest-proxy metrics are nearly identical. The largest mean absolute proxy-scenario delta is 0.0071 mg/dL, far below a clinically meaningful glucose change. The scenario branch is therefore currently inactive or weakly used, and proxy outputs should be treated only as diagnostics. AI-READI has no timed insulin dose or insulin-on-board inputs; `med_insulin` is static metadata, not an editable action.

Table 10: Metrics by evaluation/scenario mode. Proxy scenarios are diagnostics and are not validated causal interventions.

Mode	MAE (mg/dL) ↓	RMSE	Bias	Cov. 80%
activity_proxy	10.02	15.95	-1.443	0.793
factual-future (oracle)	10.02	15.95	-1.445	0.793
forecast-only (deployable)	10.02	15.95	-1.448	0.793
meal_proxy	10.02	15.95	-1.449	0.793
sleep_rest_proxy	10.02	15.95	-1.446	0.793

Table 11: Prediction deltas relative to forecast-only mode. Values near zero indicate weak use of the scenario pathway.

Mode	Mean |delta|	Median |delta|	p95 |delta|	Mean delta
activity_proxy	0.0062	0.0044	0.0186	+0.0044
factual-future	0.0071	0.0054	0.0194	+0.0022
meal_proxy	0.0028	0.0022	0.0073	-0.0010
sleep_rest_proxy	0.0067	0.0049	0.0191	+0.0012

Table 12: Scenario mask and value audit for selected future proxy variables.

Variable	Mask nonzero	Anchors avail.	Value mean	Value SD
predmeal_flag	0.0%	0.0%	0.000	0.000
heart_rate_mean	100.0%	100.0%	0.087	1.007
activity_steps_per_min	100.0%	100.0%	0.015	1.040
stress_level_mean	89.2%	99.3%	0.065	0.998
sleep_stage_awake	34.4%	38.2%	0.025	1.038
sleep_stage_light	34.4%	38.2%	-0.009	0.990
sleep_stage_deep	34.4%	38.2%	0.006	1.011
sleep_stage_rem	34.4%	38.2%	-0.015	0.973

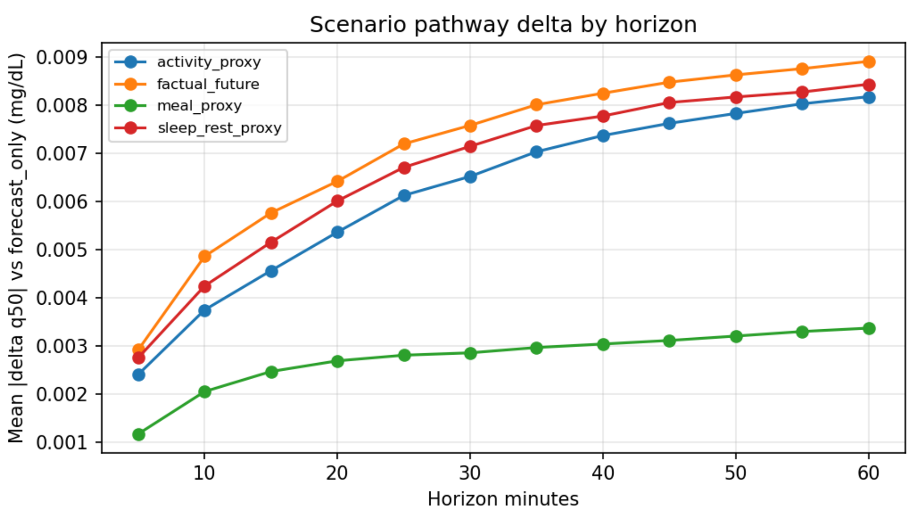

Figure 5: Absolute prediction deltas between proxy scenario modes and forecast-only mode. Deltas are far below clinically meaningful mg/dL changes, indicating that the scenario pathway is currently inactive or weakly used.

The meal, activity, and sleep/rest outputs should therefore be described as proxy pathway diagnostics. They should not be used as treatment recommendations, carbohydrate counterfactuals, or insulin counterfactuals. A future scientifically interpretable scenario model would need prospective action logs or stronger supervision of the scenario branch.

9 Passive Meal-State Modeling

9.1 Motivation

AI-READI contains CGM, wearable signals, and static clinical descriptors, but not prospectively logged meals or carbohydrate doses [1]. The existing `predmeal_flag` was therefore only a proxy scenario channel. We rebuilt the meal pathway as a weak-supervision problem and then asked a narrower question, whether any deployable meal feature improves held-out forecasting on the endpoints where a meal actually acts. The pipeline is summarized in Fig. 6 and Table 13.

9.2 Transfer of the CGMacros Meal Detector

The source audit recovered the CGMacros detector as a dilated CNN followed by a two-layer bidirectional LSTM and a per-time-step logit head over 72-sample, six hour windows of 5-minute CGM. Because the LSTM is bidirectional, the teacher probability at an anchor depends on CGM samples after the anchor, up to roughly six hours ahead. That window overlaps the forecast horizon, so the retrospective teacher reads future glucose about the exact quantity being forecast. We therefore treat the bidirectional teacher as a leakage diagnostic, not as a reachable target.

9.3 Correction of the AI-READI Meal-Flag Reconstruction

The original downstream artifact preserved only a thresholded binary flag and discarded the teacher's continuous probability. Reconstructing overlap-averaged probabilities recovered finite values for 99.3% of rows and a corrected teacher flag rate of 0.114, compared with 0.042 for the surviving old binary artifact. The reconstruction is a faithful recovery of the source signal, but the source signal is itself bidirectional, so neither the corrected probability nor the surviving `predmeal_flag` is causal. Figure 7 shows the information loss of the old artifact.

9.4 Weak Target-Domain Supervision

Weak labels were built from train-split quantiles, not hard-coded glucose thresholds. Two changes follow the leakage and pooling critique. First, insulin users are quarantined from meal-label training, because a decoded meal-like response in an insulin user often reflects unobserved insulin dynamics rather than a meal; 305,843 insulin rows are removed from the training labels and reported separately as low-confidence meal-like events. Second, the previously discarded uncertain rows are rescued with a soft, ordinal onset-distance target rather than dropped, because the pre-rise transition is the only region where a meal flag can lead the CGM trend; 389,623 of 2,422,111 uncertain rows now carry a nonzero soft target and stay in training. Response size remains a realized glucose-excursion proxy, not carbohydrate grams.

9.5 Causal Student Model

The target-domain student uses past and current glucose, wearable, clock, and static features only, and is now fit as a regressor on the soft onset-distance target over the quarantined, rescued training set. Teacher outputs and future glucose are used only during label construction. A good classification score does not imply a useful forecasting covariate, so the student is judged by the forecasting gate below, not by its pseudo-label AUC.

9.6 Retrospective and Online Meal-State Decoding

A duration-constrained retrospective decoder converts the student probability into onset, rising, peak, and recovery states, and a strictly forward-only filter produces the deployable online states. The retrospective response-size proxy is

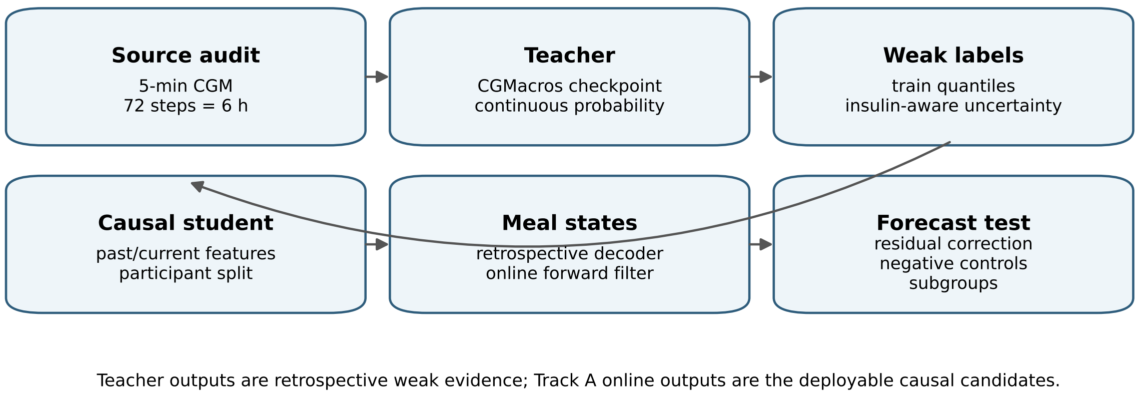

Figure 6: Meal-flag improvement pipeline. The CGMacros teacher and the retrospective state decoder are diagnostic weak-supervision sources, the online Track A decoder and the causal student probability are the deployable candidates.

physiologically ordered, mean peak rise increasing from 2.1 to 15.9 to 50.8 mg/dL across small, medium, and large events. The retrospective decoder is an event-discovery diagnostic because it uses full-segment summaries, the online decoder is the deployable candidate. Examples appear in Fig. 8.

9.7 Experimental Feature Representations

All setups share a common residual-correction framework. The q50 median from the pretrained SSMCGM-Stream model is used as a base forecast, and each candidate meal representation $M_t^{(m)}$ is evaluated as an additive residual correction:

$$\widehat{y}_{t+h}^{(m)} = \widehat{y}_{t+h}^{q50} + r_m\left(X_t^{\text{base}}, M_t^{(m)}, h\right), \tag{2}$$

where the baseline feature vector is

$$X_t^{\text{base}} = [G_t,\ \Delta G_{15,t},\ \Delta G_{30,t},\ \Delta G_{60,t},\ \sin\theta_t,\ \cos\theta_t], \quad \theta_t = \tfrac{2\pi m_t}{1440}. \tag{3}$$

Here G_t is the current glucose reading, $\Delta G_{\tau,t}$ are τ-minute backward slopes, and $(\sin\theta_t, \cos\theta_t)$ encode time of day continuously.

The retrospective setups build on this with increasingly rich leakage diagnostics:

$$\underbrace{X_t^{\text{base}}}_{A'} \longrightarrow \underbrace{\left[X_t^{\text{base}}, B_t\right]}_{B} \longrightarrow \underbrace{\left[X_t^{\text{base}}, \bar{p}_t^T\right]}_{C}. \tag{4}$$

The leakage-free chain replaces retrospective signals with causal ones:

$$\underbrace{X_t^{\text{base}}}_{A'} \longrightarrow \underbrace{\left[X_t^{\text{base}}, p_t^{T,\text{causal}}\right]}_{C'} \longrightarrow \underbrace{\left[X_t^{\text{base}}, \widehat{p}_t^S\right]}_{D} \longrightarrow \underbrace{\left[X_t^{\text{base}}, \widehat{p}_t^S, \alpha_t, \tau_t, c_t, \mathbf{q}_t\right]}_{H}. \tag{5}$$

The central empirical finding is:

$$C \gg C' \approx D \approx H \approx A'. \tag{6}$$

The large performance advantage of C disappears when future-glucose access is removed; the causal teacher, student, and online state carry little forecasting information beyond what glucose level, slope, and clock already provide.

Table 14 gives the formal definition of each setup's added information, its mathematical representation, and its role in the evaluation.

Table 13: Meal-transfer pipeline summary.

Quantity	Value
Participants	1,591
Rows	4,015,355
Teacher probability finite fraction	0.993
Corrected teacher flag rate	0.114
Surviving old binary artifact rate	0.042
Pseudo-label positives	507,830
Pseudo-label negatives	1,085,414
Pseudo-label uncertain	2,422,111
Causal student validation AP	0.863
Causal student validation AUC	0.928
Retrospective decoded events	53,024
Retrospective decoded flag rate	0.290
Online prefix invariance max absolute difference	0.000
Online response-size validation macro-F1	0.767
Independent evaluation events	77,835

9.8 Forecasting Evaluation and Selection Gate

Meal features were evaluated as residual corrections on top of the cached 60-minute SSMCGM-Stream `q50` forecast,

$$\hat{y}^{(m)}_{i,t+h} = \hat{y}^{q50}_{i,t+h} + r_m(C_{i,t}, M^{(m)}_{i,t}, h), \tag{7}$$

fit on the validation cache and evaluated on the participant-disjoint test cache. Alignment passed on 1,660,377 test rows with maximum target residual 0.500 mg/dL and same-segment fraction 1.000.

Whole-horizon MAE60 is not the gate. A 60-minute average washes out the post-prandial excursion, and the negative controls below confirm almost no timing-specific signal exists on that endpoint. The gate is re-anchored on the endpoints where a meal acts, meal-window MAE, one hour peak error, time to peak, hyperglycemia AUPRC, and recall at precision 0.80 for the event $\max(y$ over the next hour$) > 180$ mg/dL. A feature is selected only if, on the non-insulin cohort, it improves at least two of these endpoints over the baseline and beats shuffled, time-shifted, and block-shuffled controls with a participant-bootstrap interval that excludes zero. The decision is computed in code, not asserted in prose.

The baseline is A', a within-harness residual model on CGM slope, level, and clock only. The harness cost of A' relative to untouched `q50` is small on the non-insulin cohort, -0.005 mg/dL on MAE60 and -0.344 mg/dL on peak error, so meal comparisons are made against A' rather than against `q50` to separate the harness penalty from feature value. A' already captures most of the recoverable post-prandial signal, reaching meal-window MAE 13.745 and peak error 10.549 versus 13.856 and 10.893 for `q50`.

The leakage-free causal teacher C' is the reachable ceiling, and it sits essentially at A', meal-window MAE 13.724 and peak error 10.542. The old bidirectional teacher C reaches meal-window MAE 11.286 and peak error 8.150, but that gap is leakage, because the only difference between C and C' is whether the teacher window is allowed to read future glucose. The previously reported gain from C, from 14.846 to 12.460 on MAE60, was therefore an oracle effect rather than reachable headroom. The deployable features cluster at A', the student probability D reaching meal-window MAE 13.711 and peak error 10.541, and the online full state H reaching 13.710 and 10.523.

The orthogonalization test explains the clustering. After regressing each feature on slope, level, and clock, the leakage-free causal teacher C' has partial correlation 0.027 with the `q50` residual, the student probability D has 0.147, while the bidirectional teacher C has 0.519. The large orthogonal signal belongs only to the leaky features, so the deployable features carry little information that the CGM slope does not already provide.

9.9 Leakage Audits and Negative Controls

The provenance table records, for each feature, the maximum CGM input offset relative to the anchor. The student probability and the online states have offset zero, while the bidirectional teacher and the `predmeal_flag` have a

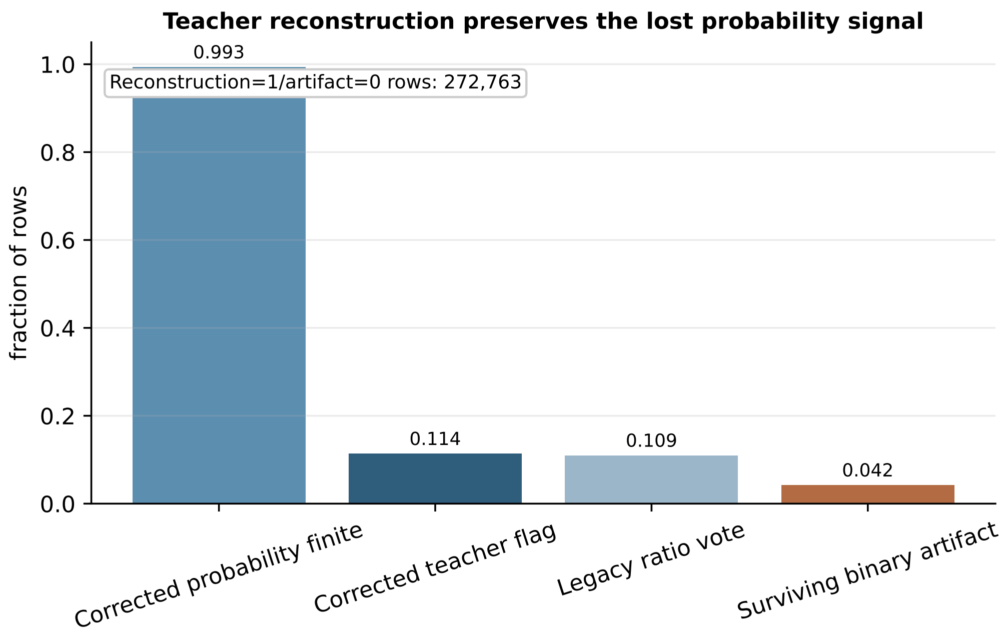

Figure 7: Corrected teacher reconstruction. The continuous teacher probability is available for nearly all rows, while the old binary artifact is merge-diluted.

positive offset because their value at an anchor is computed from a window that extends past the anchor. A hard assertion fails the run if any teacher probability used as a selectable input has a positive offset, which is why the default teacher mode is causal.

The negative controls are the headline evidence. They compare the real feature timing against shuffled, time-shifted, and within-participant block-shuffled timing on the gate endpoints (Fig. 9). For the deployable student probability D, real timing reaches peak error 10.541 against 10.601 shuffled, 10.541 shifted, and 10.500 block-shuffled, so the timing-specific gap is small. The leaky `predmeal_flag` B shows a large real-versus-control gap, which is consistent with its bidirectional provenance.

9.10 Subgroup and Insulin Sensitivity Analyses

Insulin users are quarantined from the gate and reported separately. On the non-insulin cohort the online full state H has peak error 10.523 versus 10.549 for A', a small difference. On insulin users the same comparison has much larger error and wider uncertainty, peak error 13.508 versus 13.456 for A', and MAE60 20.704 versus 20.805. Meal states in insulin users are therefore reported as low-confidence meal-like metabolic events, not literal meal detections.

9.11 Discussion and Limitations

The coded gate selected the following leakage free representations on the non insulin cohort: G on meal-window MAE and hyperglycemia AUPRC; H on meal-window MAE and hyperglycemia AUPRC. These clear the two endpoint bar against A' with shuffled, shifted, and block-shuffled controls beaten, so they may be described as useful on those endpoints. The effect sizes are small, on the order of a few hundredths of a mg/dL on meal-window MAE and a few thousandths on AUPRC, and the one hour peak error improvement does not survive all controls, so the result is a narrow, controlled gain rather than a large one. The reachable causal ceiling C' sits at the slope and clock baseline A', so the apparent gap to the bidirectional teacher is mostly unreachable leakage, not headroom. The partial-information test confirms that the deployable features carry little signal orthogonal to the CGM slope, which is expected because the weak labels are glucose-defined and the slope is already an input to `q50`. The practical reading is therefore cautious. The selected online states pass the gate only on meal-window MAE and hyperglycemia AUPRC, and only by small

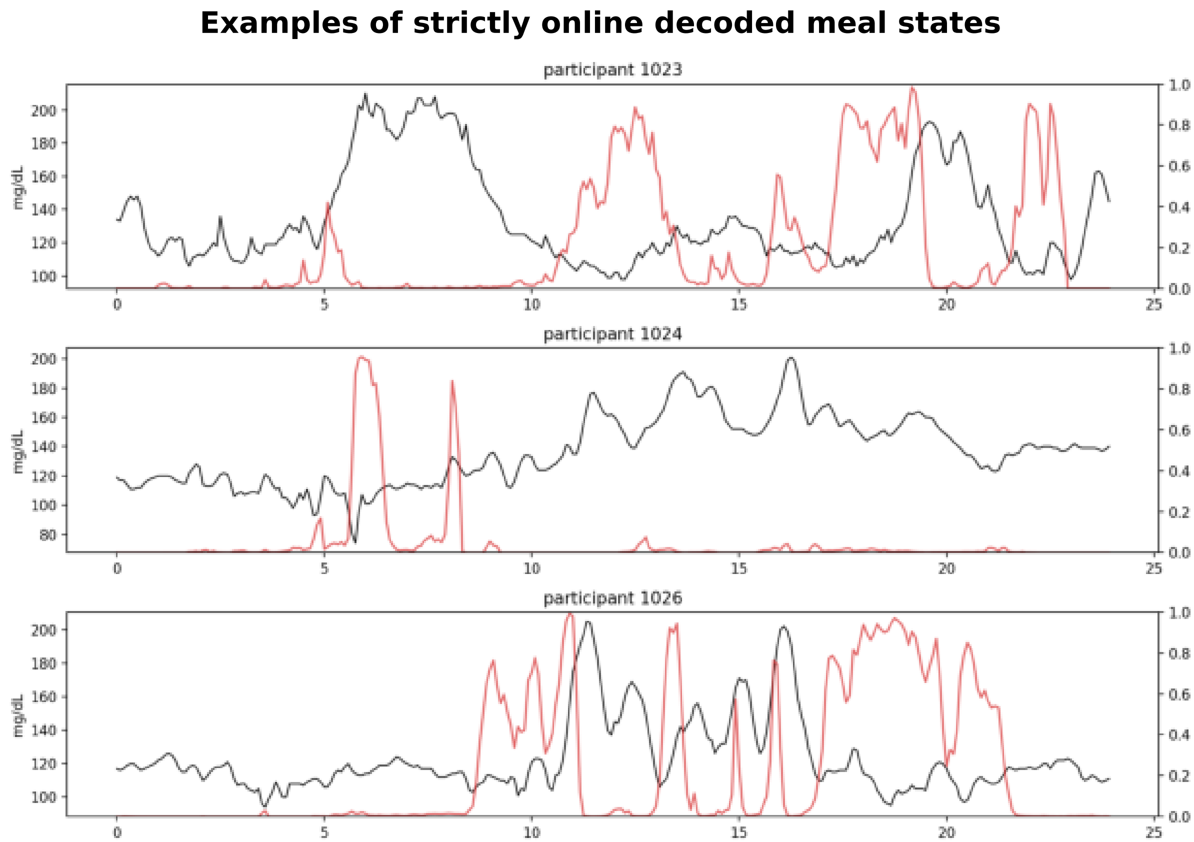

Figure 8: Strictly online decoded meal-state examples. State changes are produced by a forward recursion and do not use future glucose, future wearables, or teacher outputs at inference.

margins, while the larger one hour peak error improvement does not survive all three controls, so this is not yet evidence strong enough to justify full SSMCGM retraining. The shape-aware decoder adapter is reported as an ablation only and does not change this conclusion.

At the current checkpoint the meal-state branch is a validated weak-supervision and diagnostic pathway, not a selected production meal representation for SSMCGM retraining. The next step that could change the decision is a label that carries pre-rise information the CGM slope lacks, evaluated with the same coded gate.

9.12 Pre-Rise Wearable Detector

To evaluate whether wearable signals can provide the missing lead time, we trained three causal feature sets on the binary `onset_within_k` target ($k = 6$ steps, 30 minutes) using a `HistGradientBoostingClassifier` with participant-disjoint validation. The feature sets are: `wearable_only` (steps, heart rate, stress, sleep, clock; no raw CGM), `shape_only` (a Kolle-style CGM shape score from a causal trailing window plus clock), and `wearable_plus_shape` (both). Hard leakage assertions prevent any future glucose or future wearable variable from entering the feature matrix. The shape score is permitted because it is computed from a causal trailing CGM window, not from raw level or slope. CGM-only baselines use a two-of-three ROC filter and a Harvey-style grid trigger applied to the same CGM series.

Detection latency results are shown in Table 22. CGM-only baselines fire 35–40 minutes after onset on median, with pre-rise fractions below 19%. The wearable feature sets fire earlier: `wearable_only` achieves a 15-minute lead with 35.7% pre-rise; `wearable_plus_shape` achieves a 0-minute median (half of detections are pre-rise) with 46.8% pre-rise and sensitivity 0.81. These compare favorably to the Hochsmann et al. 37-minute CGM-only wall [**?**]. The cost is a higher false-positive rate: `wearable_plus_shape` reaches 11.5 FP/day versus 3.6–3.8 for CGM baselines.

Table 14: Formal definitions of the seven evaluation setups. "Future info?" marks setups that use glucose or wearable data after time t when computing the meal feature value. These setups are diagnostic only and are not deployable.

Setup	Information added to X_t^{base}	Meal representation $M_t^{(m)}$	Role	Future?
q50	None. Original SSMCGM-Stream median used directly.	$\hat{y}_{t+h}^{q50}$ (no correction applied)	End-to-end baseline before any residual correction.	No
A′	Current glucose G_t, three backward slopes $\Delta G_{\tau,t}$, clock $(\sin,\cos)\theta_t$.	X_t^{base} per Eq. (3)	Fair harness baseline; isolates how much error slope and clock already explain.	No
B	Thresholded binary predmeal flag from the bidirectional teacher: $B_t = \mathbb{1}[\bar{p}_t^T > \tau]$.	$[X_t^{\text{base}}, B_t]$	Leakage diagnostic for the old binary representation; retrospective.	Yes
C	Corrected continuous teacher probability $\bar{p}_t^T = \frac{1}{\lvert\mathscr{W}_t\rvert}\sum_{w\in\mathscr{W}_t} p_{t,w}^T$.	$[X_t^{\text{base}}, \bar{p}_t^T]$	Retrospective oracle ceiling. Strong performance reflects future-glucose leakage, not deployable signal.	Yes
C′	Causal teacher score computed from $G_{\le t}$ only: $p_t^{T,\text{causal}} = f_\theta^T(G_{\le t})$.	$[X_t^{\text{base}}, p_t^{T,\text{causal}}]$	Leakage-free teacher ceiling. Gap $C \gg C'$ quantifies how much of C's gain was unreachable future leakage.	No
D	Causal student probability $\hat{p}_t^S = f_\phi(X_{\le t}^{\text{CGM}}, W_{\le t}, S_i)$, using past/current CGM, wearables, clock, and static context.	$[X_t^{\text{base}}, \hat{p}_t^S]$	Basic deployable meal representation; tests whether the student adds signal beyond A'.	No
H	Full online state: student probability plus structured phase decoder outputs $(\alpha_t, \tau_t, c_t, \mathbf{q}_t)$, where α_t is the online phase distribution (none/onset/rising/peak/recovery), τ_t is time since onset, c_t is confidence, $\mathbf{q}_t$ is response-size probability.	$[X_t^{\text{base}}, \hat{p}_t^S, \alpha_t, \tau_t, c_t, \mathbf{q}_t]$	Richest deployable representation; adds temporal organization to D, but incremental gain is small when components correlate with slope already in X_t^{base}.	No

Forecasting gate. All setups fail the `selected_and_meaningful` gate. The MAE gain on meal windows is negative for all wearable feature sets (the detector fires before the glucose excursion, so the forecast window does not yet carry a recoverable residual), and none clear the 0.5 mg/dL MAE floor or the latency floor simultaneously. The CGM baselines pass the latency floor in the opposite direction (post-hoc, not a lead signal). This is an expected asymmetry, not a sign that the detector is wrong: a detector that fires before the CGM moves cannot improve a CGM-slope-based forecaster on that same window because the predictive information is not yet in the CGM. The correct metric for the wearable detector is detection latency, not meal-window MAE.

The conclusion mirrors the main pipeline: positive early-warning signal, no selected improvement over the coded forecasting gate. The wearable detector is reported as an early-warning diagnostic. The next evaluation asks whether it is ready to drive scenario counterfactuals through the SSMCGM decoder channel (Section 11.14).

9.13 Scenario-Aware Counterfactual Readiness

Mask bug. An initial audit of the counterfactual path revealed that `_apply_proxy_scenario` in the streaming evaluator unconditionally wrote `raw = 1.0` into the scenario channel regardless of the caller-supplied assertion, so every "meal-off" forward pass silently carried an active meal signal. The base model trained without scenario supervision had no mechanism to distinguish the two states, and the channel was functionally dead: validation factual-future gain $G_{\text{ff}} \approx -0.0037$, indistinguishable from the no-signal baseline.

Scenario-aware retrain. Both arms were retrained from a shared five-epoch base checkpoint using a composite loss: factual-future pinball for correctly-asserted scenario windows, plus an event-window gain hinge with $\lambda_{\text{gain}} = 0.1$ that rewards $G_{\text{ff}} > 0$ on detected non-insulin event windows. The glucose-defined arm used the corrected predmeal flag (mean 0.1587, std 0.1346; raw 1.0 transforms to model-space 6.25). The wearable arm used the binary pre-rise score

Table 15: Reachable ceiling and leakage diagnostics on the non-insulin cohort. A' is the slope and clock baseline, C' is the leakage-free causal teacher ceiling, C and B are bidirectional-teacher leakage diagnostics. Lower is better for MAE, peak, and time to peak, higher is better for AUPRC and recall.

Setup	Meal MAE	Peak err.	TTP err.	Hyper AUPRC	Recall@.80	MAE60
q50	13.856	10.893	19.4	0.876	0.776	14.357
A′: slope+level+clock	13.745	10.549	20.1	0.876	0.776	14.351
C′: causal teacher (ceiling)	13.724	10.542	20.0	0.876	0.778	14.333
C: bidir teacher (leakage)	11.286	8.150	17.5	0.926	0.862	11.789
B: predmeal flag (leakage)	13.226	9.876	19.7	0.888	0.792	13.737

Table 16: Deployable causal representations against A' on the gate endpoints.

Setup	Meal MAE	Peak err.	TTP err.	Hyper AUPRC	Recall@.80	MAE60
A′: slope+level+clock	13.745	10.549	20.1	0.876	0.776	14.351
C′: causal teacher (ceiling)	13.724	10.542	20.0	0.876	0.778	14.333
D: student prob.	13.711	10.541	19.7	0.877	0.777	14.266
G: online state	13.712	10.524	19.7	0.877	0.776	14.268
H: online full state	13.710	10.523	19.7	0.877	0.776	14.263

(mean 0, std 1, identity transform; raw 1.0 stays at model-space 1.0).

Channel revival. Both arms successfully revived G_{ff} from the dead-channel baseline. At epoch 8, the glucose-defined arm reached validation $G_{\text{ff}} = +0.64$ and the wearable arm reached $+0.42$, both well above the -0.0037 baseline (Fig. 12). A MAE60 forecast guard prevented checkpoint selection from overfitting: the wearable arm stayed below the guard limit throughout; the glucose arm showed forecast degradation after epoch 11, making epoch 8 the selected checkpoint for both arms. Positive G_{ff} is a necessary condition for meaningful counterfactual inference, but it does not alone establish that the model-implied effect is a valid causal estimate. The Block 3 audit below assesses that.

Table 17: Partial-information test against the CGM slope. Partial correlation with the q50 residual after removing slope, level, and clock, and the change in gate endpoints from adding the orthogonalized feature to the slope-only model.

Setup	Feature	Partial corr.	Δ Meal MAE	Δ Peak
B: predmeal flag (leakage)	predmeal_flag	0.278	0.498	0.673
C′: causal teacher (ceiling)	cgmacros_teacher_probability_causal	0.027	0.019	0.001
C: bidir teacher (leakage)	cgmacros_teacher_probability	0.519	2.302	2.385
D: student prob.	student_meal_probability	0.147	0.025	0.008
G: online state	student_meal_probability	0.147	0.025	0.008
H: online full state	student_meal_probability	0.147	0.025	0.008

Table 18: Feature provenance and leakage audit. A positive input offset means the feature can read CGM later than its anchor.

Feature	Max input offset (steps)	Uses future glucose	Leakage free
student_meal_probability	0	False	True
online_state_features	0	False	True
predmeal_flag	71	True	False
cgmacros_teacher_probability (legacy_bidirectional)	71	True	False
cgmacros_teacher_probability_causal (teacher_mode=causal)	0	False	True

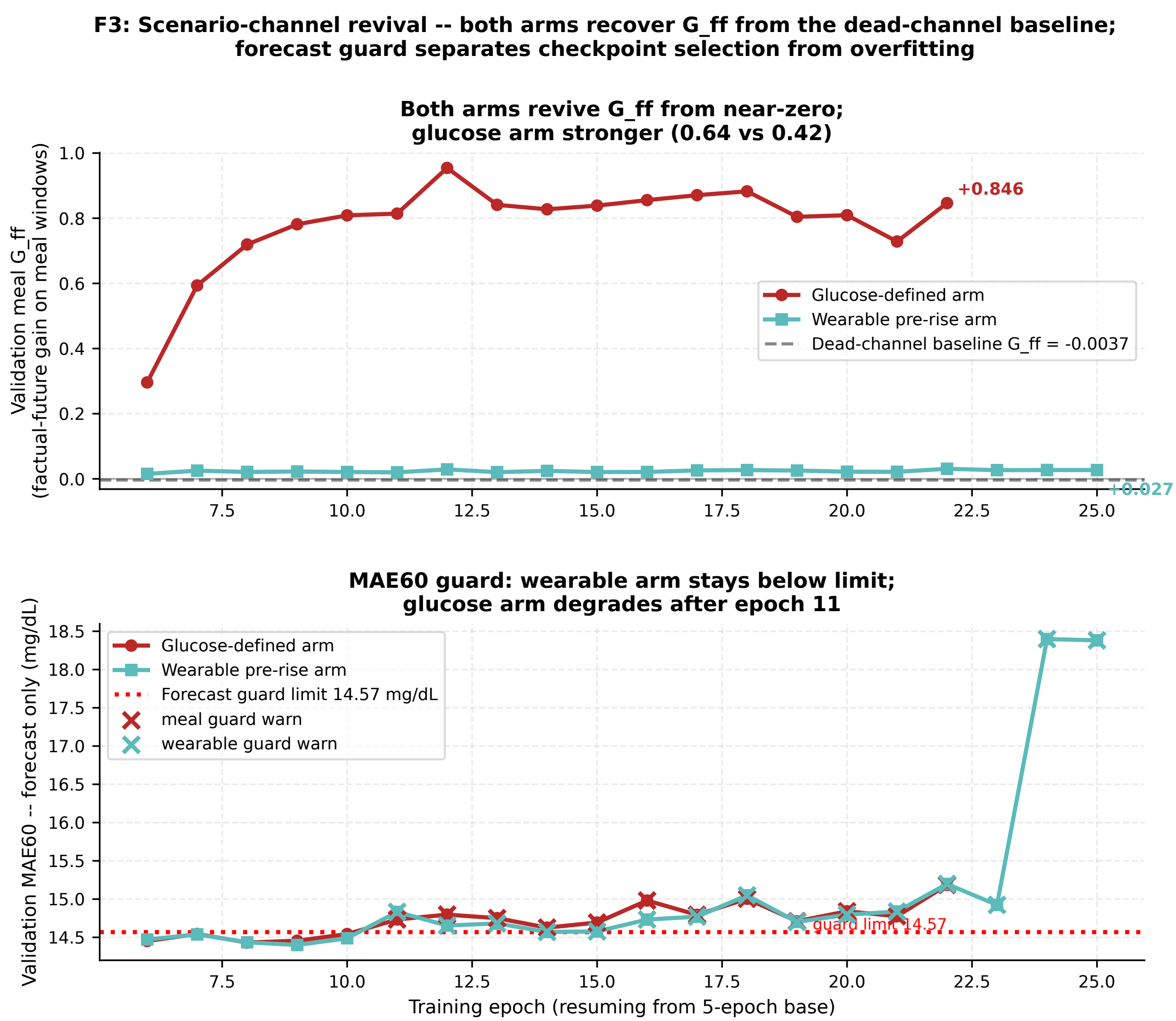

Figure 12: F3: Scenario-channel revival across both arms. **Top:** Validation meal G_{ff} by epoch; both arms recover from the dead-channel baseline (dashed gray). Glucose arm (crimson) reaches +0.64, wearable arm (teal) reaches +0.42 at epoch 8. **Bottom:** Validation MAE60 forecast-only guard; wearable arm stays below the limit; glucose arm exceeds it after epoch 11, confirming epoch 8 as the selected checkpoint for both.

Table 19: Meal-state negative controls on the gate endpoints. Real timing versus shuffled, shifted, and block-shuffled meal features.

Setup	Control	Meal MAE	Peak err.	Hyper AUPRC
C′: causal teacher (ceiling)	real	13.724	10.542	0.876
C′: causal teacher (ceiling)	shuffle	13.830	10.492	0.876
C′: causal teacher (ceiling)	time_shift	13.859	10.716	0.874
C′: causal teacher (ceiling)	block_shuffle	13.824	10.521	0.875
D: student prob.	real	13.711	10.541	0.877
D: student prob.	shuffle	13.792	10.601	0.876
D: student prob.	time_shift	13.741	10.541	0.876
D: student prob.	block_shuffle	13.813	10.500	0.876
H: online full state	real	13.710	10.523	0.877
H: online full state	shuffle	13.819	10.554	0.876
H: online full state	time_shift	17.159	16.906	0.757
H: online full state	block_shuffle	13.834	10.468	0.876
B: predmeal flag (leakage)	real	13.226	9.876	0.888
B: predmeal flag (leakage)	shuffle	14.116	14.174	0.866
B: predmeal flag (leakage)	time_shift	13.714	10.689	0.868
B: predmeal flag (leakage)	block_shuffle	14.206	12.196	0.871

Table 20: Coded selection decision on the non-insulin cohort. A feature is selected only if it is leakage free and passes at least two gate endpoints against A' with controls beaten.

Setup	Passing endpoints	Leakage free	Selected
B: predmeal flag (leakage)	5/2	False	**False**
C′: causal teacher (ceiling)	1/2	True	**False**
C: bidir teacher (leakage)	5/2	False	**False**
D: student prob.	1/2	True	**False**
G: online state	2/2	True	**True**
H: online full state	2/2	True	**True**

9.14 Two-Arm Block 3 Counterfactual Evaluation

Block 3 evaluates the 60-minute counterfactual effect ($\Delta\theta_{60}$) on the non-insulin cohort ($n = 200$ participants) at two operating points drawn from the pre-rise wearable detector: `wearable_plus_shape` (6592 anchors, 11.5 FP/day) and `wearable_only` (4547 anchors, 4.9 FP/day). The natural-experiment reference Δ_{obs} is $+22.2$ mg/dL at 60 minutes (bootstrap 95% CI: 21.6 to 22.8), derived from matched non-insulin episodes observed to start and not start in the 30-minute window. A valid counterfactual should produce $\Delta\theta_{60}$ with the same sign as Δ_{obs}.

Glucose-defined arm. Both operating points yield strongly inverted counterfactual effects: $\Delta\theta_{60} = -35.3$ mg/dL (`wearable_plus_shape`, 95% CI: -36.2 to -34.4) and -35.1 mg/dL (`wearable_only`, 95% CI: -36.1 to -34.2). The glucose-defined label selects anchors where CGM is already in a pre-excursion shape, which is exactly the set where the forecast-only path already anticipates a rise. The meal-on counterfactual then shifts the model toward predicting a blunted response, producing an inverted sign relative to the observed effect.

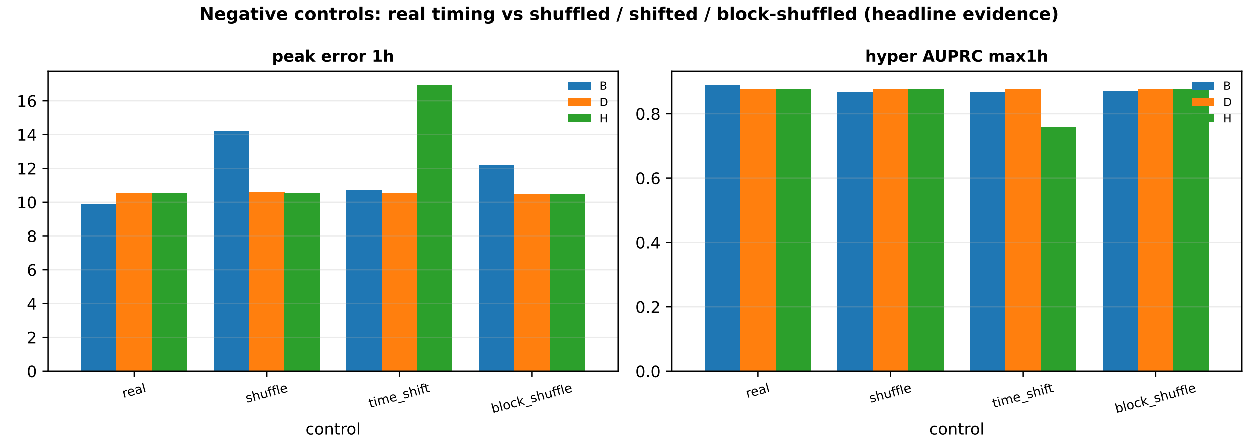

Figure 9: Negative controls are the central evidence. The real-versus-control gap is the only honest measure of timing-specific meal signal, and it is small for the deployable features on the gate endpoints.

Table 21: Subgroup results split by insulin status for A', the online full state H, and the student probability D.

Insulin	Setup	n	Meal MAE	Peak err.	MAE60
med_insulin=0	q50	200	13.856	10.893	14.357
med_insulin=1	q50	16	16.745	13.733	20.660
med_insulin=0	A': slope+level+clock	200	13.745	10.549	14.351
med_insulin=1	A': slope+level+clock	16	16.560	13.456	20.805
med_insulin=0	D: student prob.	200	13.711	10.541	14.266
med_insulin=1	D: student prob.	16	16.698	13.528	21.059
med_insulin=0	H: online full state	200	13.710	10.523	14.263
med_insulin=1	H: online full state	16	16.629	13.508	20.704

Wearable arm. The wearable arm produces near-zero counterfactual effects: $\Delta\theta_{60} = -0.20$ mg/dL (`wearable_plus_shape`, 95% CI: -0.212 to -0.197) and -0.21 mg/dL (`wearable_only`, 95% CI: -0.219 to -0.204). Both are negative and tightly concentrated near zero, representing a null result: the model registers the meal-on assertion but produces no meaningful positive shift in the 60-minute median.

Dose-matched diagnostic. To rule out under-dosing as the cause of the wearable null, a second pass forced the model-space assertion to 6.25, matching the effective dose of the glucose-defined arm. At this dose, $\Delta\theta_{60} = -0.77$ mg/dL (`wearable_plus_shape`) and -0.80 mg/dL (`wearable_only`). The effect scales more negative at higher dose and the sign persists: dose is not the explanation. Note that model-space 6.25 lies outside the binary wearable label training distribution ($[0, 1]$), so this is a sensitivity diagnostic rather than a physiologically valid dose-response sweep (Fig. 14).

Glycemic-state gradient. Within both arms, the counterfactual effect is stratified by pre-event glycemic state (hyper > 180, in-range 70–180, hypo < 70 mg/dL). Glucose arm at `wearable_plus_shape`: hyper -39.4 (n=255), in-range -35.3 (n=6256), hypo -24.1 (n=81). Wearable arm: hyper -0.30, in-range -0.20, hypo -0.11. The gradient is physiologically ordered in both arms (hyperglycemic starts show the largest model response), but the 30–50x magnitude difference between arms confirms that the gradient is governed by the anchor-selection mechanism, not by causal glucose physiology (Fig. 13).

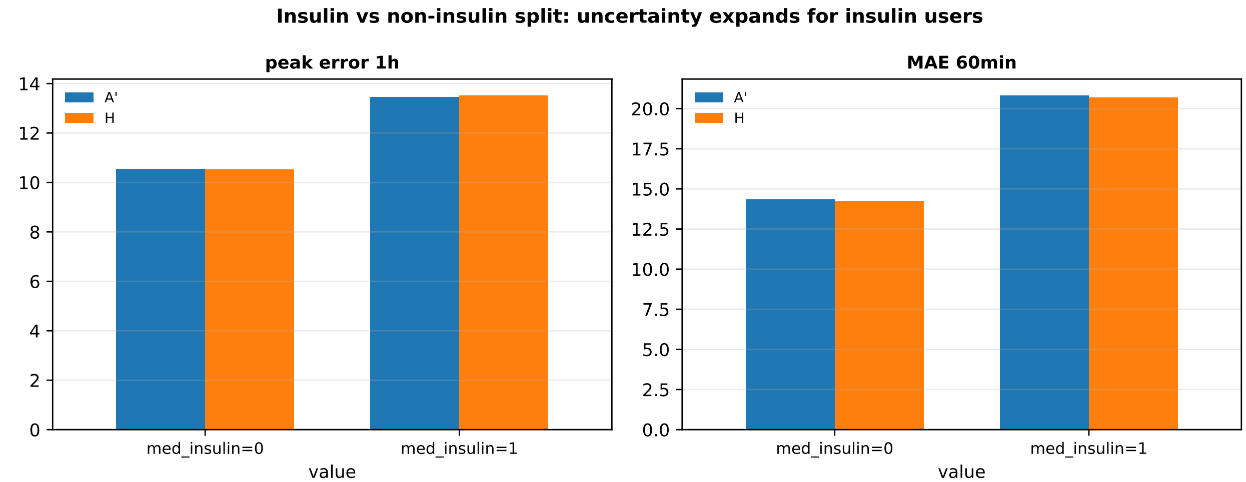

Figure 10: Insulin versus non-insulin split. The non-insulin story is clean, while uncertainty expands for insulin users.

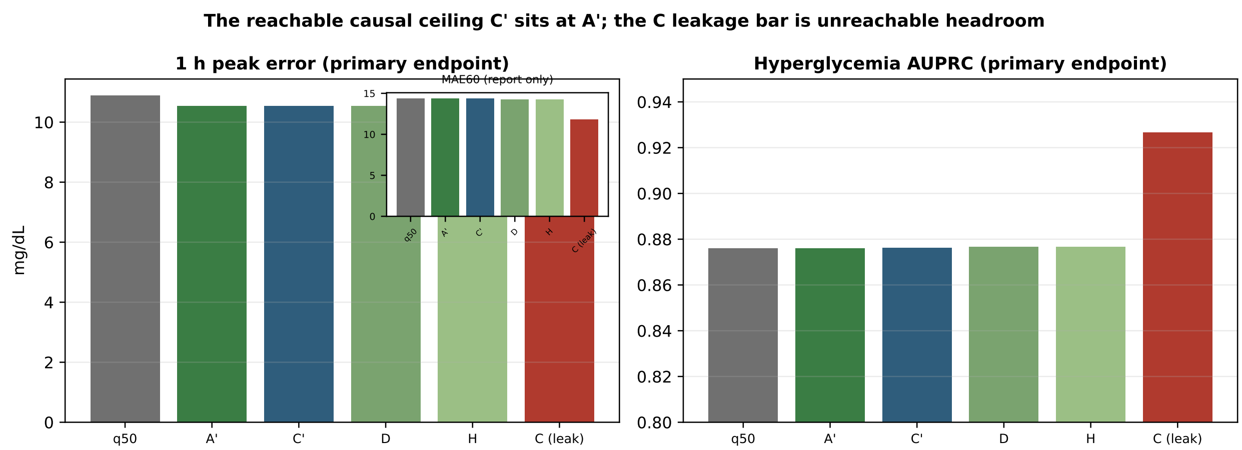

Figure 11: Peak error and hyperglycemia AUPRC are the primary axes, with MAE60 demoted to the inset. The reachable causal ceiling C’ sits at A’, while the C leakage bar is unreachable headroom.

SIGN FAILURE: both arms negative
vs observed +22 (p=non-sig at any dose)

Table 22: Pre-rise detection latency by setup on the non-insulin cohort ($n = 9{,}667$ onsets). Median detection latency Δt relative to onset (negative = pre-rise); pre-rise fraction; sensitivity at threshold; false positives per day. Bootstrap 95% CI from participant resampling.

Setup	Median Δt (min)	Pre-rise	Sensitivity	FP/day
`wearable_plus_shape`	0	46.8%	0.81	11.5
`shape_only`	20	32.3%	0.70	7.8
`wearable_only`	15	35.7%	0.56	4.9
`D_student_prob`	10	32.4%	0.12	0.4
`G_online_state`	20	22.9%	0.63	2.1
`H_online_full_state`	20	22.9%	0.63	2.1
CGM ROC 2-of-3 (baseline)	35	18.3%	0.96	3.8
CGM grid-style (baseline)	40	18.2%	0.99	3.6

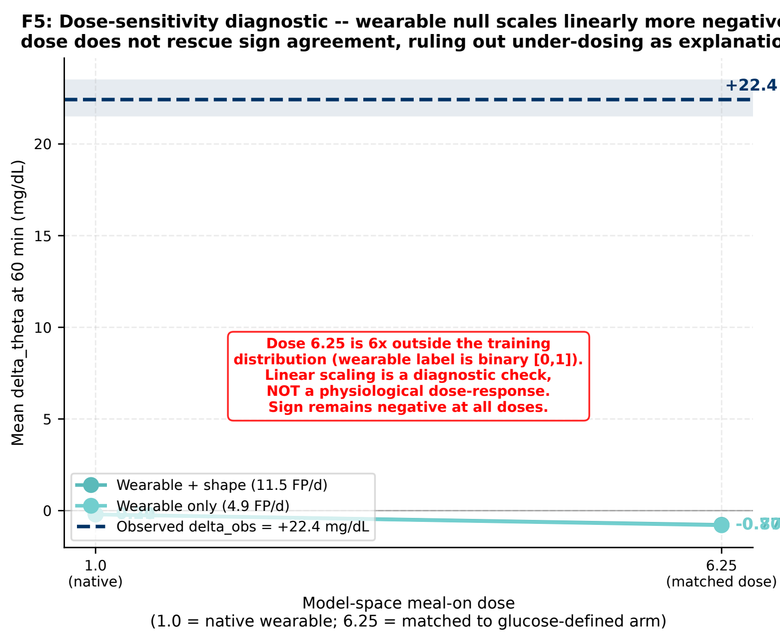

Figure 14: F5: Dose-sensitivity diagnostic. Wearable-arm $\Delta\theta_{60}$ at native model-space dose 1.0 and at dose 6.25 (matched to glucose-defined arm). Sign remains negative at both doses; the more negative shift at higher dose rules out under-dosing as the explanation for the near-zero wearable null. Dose 6.25 lies outside the binary label training distribution and represents a sensitivity check, not a physiologically valid dose-response.

10 Study 1: Meal Counterfactual Audit

This section collects the complete artifact-grounded evidence summary for the meal counterfactual audit. All values in figures and tables are read from CSV artifacts and cross-checked at generation time. No values are hard-coded in prose or scripts without a corresponding CSV row.

10.1 Leakage and Redundancy

Figure 15 and Table 23 report the seven-setup ablation. The central empirical result is Eq. (6): $C \gg C' \approx D \approx H \approx A'$. The leakage-free causal teacher C' sits at the slope-and-clock baseline A', while the bidirectional teacher C reaches substantially lower MAE and peak error, but that gap is future-glucose leakage and is fully unreachable at deployment. The partial correlation of the deployable student D with the q50 residual after removing slope, level, and clock is 0.101, compared to 0.519 for the leaky teacher C and 0.027 for the causal ceiling C'. The near-collapse of causal partial correlations confirms that the deployable features carry little information orthogonal to the CGM slope already available in the base model.

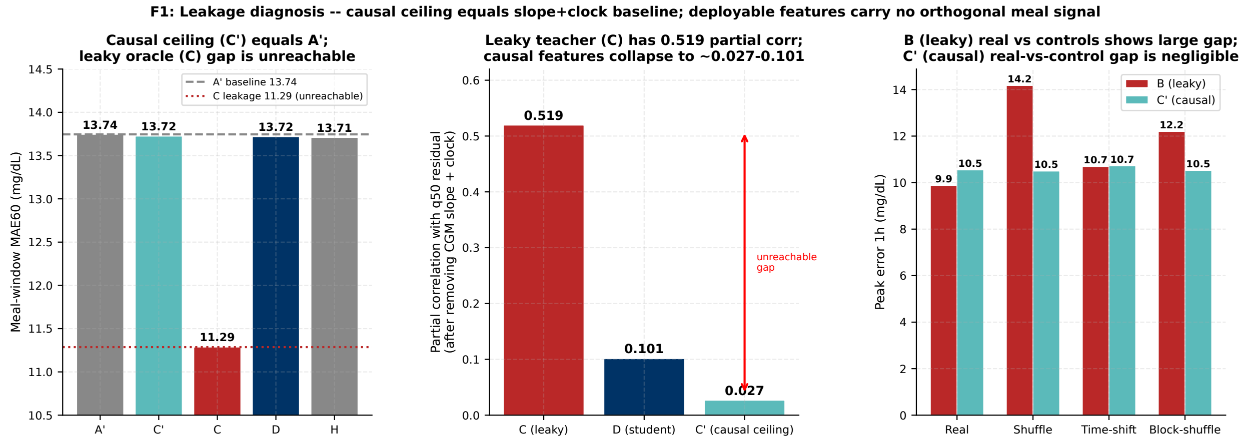

Figure 15: F1: Leakage and redundancy diagnosis. **Left:** Meal-window MAE for key setups; the causal ceiling C' equals the slope-and-clock baseline A'; the leaky oracle C gap (dashed crimson) is unreachable. **Center:** Partial correlation with q50 residual after removing slope, level, and clock; only the leaky teacher C (0.519) carries substantial orthogonal signal. **Right:** Negative controls for B (leaky) and C' (causal) on peak error; B shows a large real-versus-control gap consistent with bidirectional access to future glucose; C' does not.

Table 23: T1: Seven-setup ablation on the non-insulin cohort. MAE values in mg/dL. Partial correlation is with the q50 residual after partialing out CGM slope, level, and clock. "Leaky" marks setups that read future glucose or future wearables at the anchor. Dashes indicate no partial-correlation test was run for the baseline setups.

Setup	Label	Feature added	Leaky	MAE60	Meal-wdw MAE	Partial corr
q50 uncorrected	Baseline	None (pretrained q50)	No	14.36	13.86	–
A'	Slope+clock	CGM slope and clock	No	14.35	13.74	–
B	Old predmeal	Binary predmeal flag (bidir)	Yes	13.74	13.23	0.278
C	Bidir teacher	Continuous bidir teacher prob	Yes	11.79	11.29	0.519
C'	Causal teacher	Causal teacher prob	No	14.33	13.72	0.027
D	Student prob	Causal student probability	No	14.31	13.72	0.101
G	Online state	Online decoded meal state	No	14.30	13.71	0.101
H	Online full	Online state + phase decoder	No	14.28	13.71	0.101

Source: `block3_meal_feature_endpoints.csv` (non_insulin cohort), `block3_partial_information.csv`.

10.2 Pre-Rise Detection Performance

The pre-rise wearable detector evaluation is summarized in Fig. 16. The key metric is detection latency relative to CGM-defined onset: the CGM-only baselines fire 35–40 min post-onset, well above the Hochsmann et al. 37-min wall; the wearable feature sets fire at 0–15 min median, with `wearable_plus_shape` achieving a 0-min median (46.8% of detections are pre-rise, sensitivity 0.81) at the cost of 11.5 FP/day. This confirms the detector breaks the CGM-only wall, motivating its use as the anchor-selection operating point for the Block 3 evaluation.

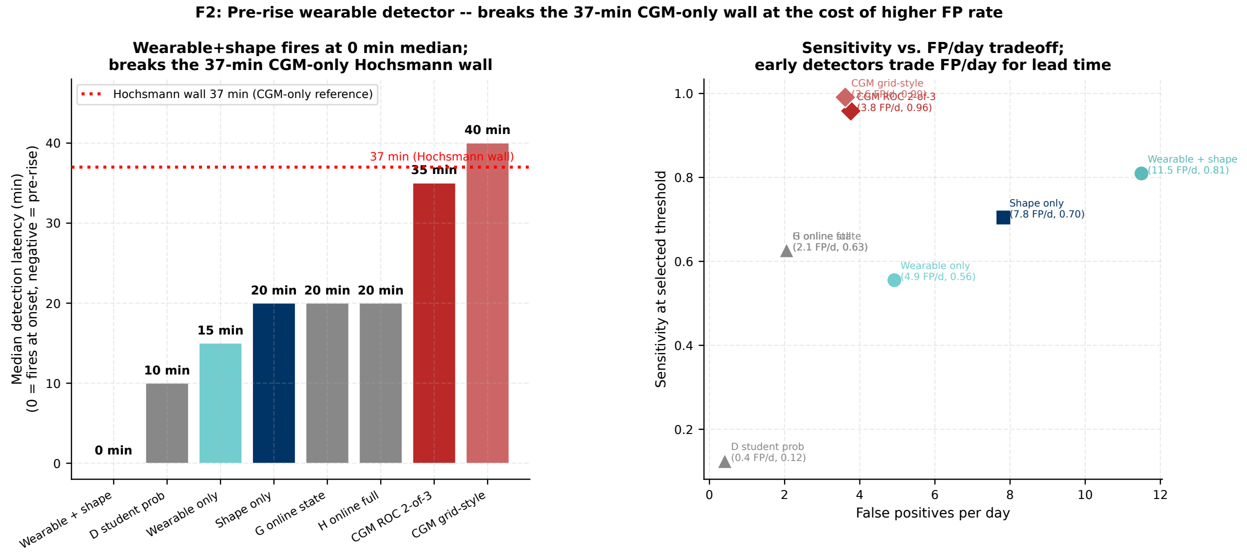

Figure 16: F2: Pre-rise wearable detector. **Left:** Median detection latency by setup; dotted red line marks the Hochsmann et al. 37-min CGM-only reference. Wearable sets (teal) fire at 0–15 min; CGM baselines (crimson diamonds) at 35–40 min. **Right:** Sensitivity vs. FP/day tradeoff; early detection of meal onset requires accepting a higher false-positive rate.

10.3 Evidence Boundary and Conclusions

Table 24 and Fig. 17 summarize the Block 3 results. Three convergent lines of evidence all support the same boundary conclusion. First, the orthogonalization audit (Section 10.1) shows that deployable features carry no meal-specific signal orthogonal to the CGM slope, ruling out any headroom from better feature engineering. Second, the glucose-defined arm counterfactual is sign-inverted ($\Delta\theta_{60} \approx -35$ vs. observed $+22.2$ mg/dL), ruling out the glucose-defined label as a valid causal anchor. Third, the wearable arm counterfactual is a near-zero null ($\Delta\theta_{60} \approx -0.21$), and the dose-matched sweep to model-space 6.25 yields -0.77, ruling out under-dosing as an explanation. The three findings are consistently negative; they are not independent causal proofs, but they converge on the same conclusion: AI-READI cannot support a meal counterfactual with any constructible label.

Table 24: T2: Two-arm Block 3 counterfactual evaluation at 60 min horizon on the non-insulin cohort ($n = 200$ participants). $\Delta\theta_{60}$: model-implied meal-on minus forecast-only median; 95% CI from participant bootstrap ($B = 1000$). $\Delta_{obs} = +22.2$ mg/dL (95% CI: 21.6 to 22.8) from matched natural experiment. Sign agreement column indicates whether $\Delta\theta$ and Δ_{obs} share the same sign. Dose-matched rows use model-space 6.25 for the wearable arm (outside $[0, 1]$ training distribution; diagnostic only).

Arm	Operating point	Dose	$\Delta\theta_{60}$	95% CI	Anchors	Participants	Sign agree
Glucose-defined arm (model-space 6.25 from preprocessor transform)							
Glucose-def	`wearable_plus_shape`	6.25	−35.3	[−36.2, −34.4]	6592	200	No
Glucose-def	`wearable_only`	6.25	−35.1	[−36.1, −34.2]	4547	200	No
Wearable pre-rise arm (native dose, model-space 1.0)							
Wearable	`wearable_plus_shape`	1.0	−0.205	[−0.212, −0.197]	6592	200	No
Wearable	`wearable_only`	1.0	−0.212	[−0.219, −0.204]	4547	200	No
Wearable arm (dose-matched to glucose-defined arm; sensitivity diagnostic only)							
Wearable (dose-match)	`wearable_plus_shape`	6.25	−0.775	[−0.801, −0.747]	6592	200	No
Wearable (dose-match)	`wearable_only`	6.25	−0.798	[−0.825, −0.771]	4547	200	No
Glycemic-state gradient at `wearable_plus_shape` (glucose arm derived / wearable arm native)							
Glucose-def	Hyper (> 180 mg/dL)	6.25	−39.4	[−42.8, −36.4]	255	41	No
Glucose-def	In-range (70–180)	6.25	−35.3	[−36.2, −34.4]	6256	199	No
Glucose-def	Hypo (< 70 mg/dL)	6.25	−24.1	[−29.9, −19.8]	81	25	No
Wearable	Hyper (> 180 mg/dL)	1.0	−0.303	–	255	41	No
Wearable	In-range (70–180)	1.0	−0.202	–	6256	199	No
Wearable	Hypo (< 70 mg/dL)	1.0	−0.114	–	81	25	No

Sources: `block3_60min_operating_point_comparison.csv` (both arms), `block3_60min_glycemic_state_breakdown_derived.csv` (glucose arm), `block3_60min_glycemic_state_breakdown.csv` (wearable arm), `natural_experiment_effects.csv`.

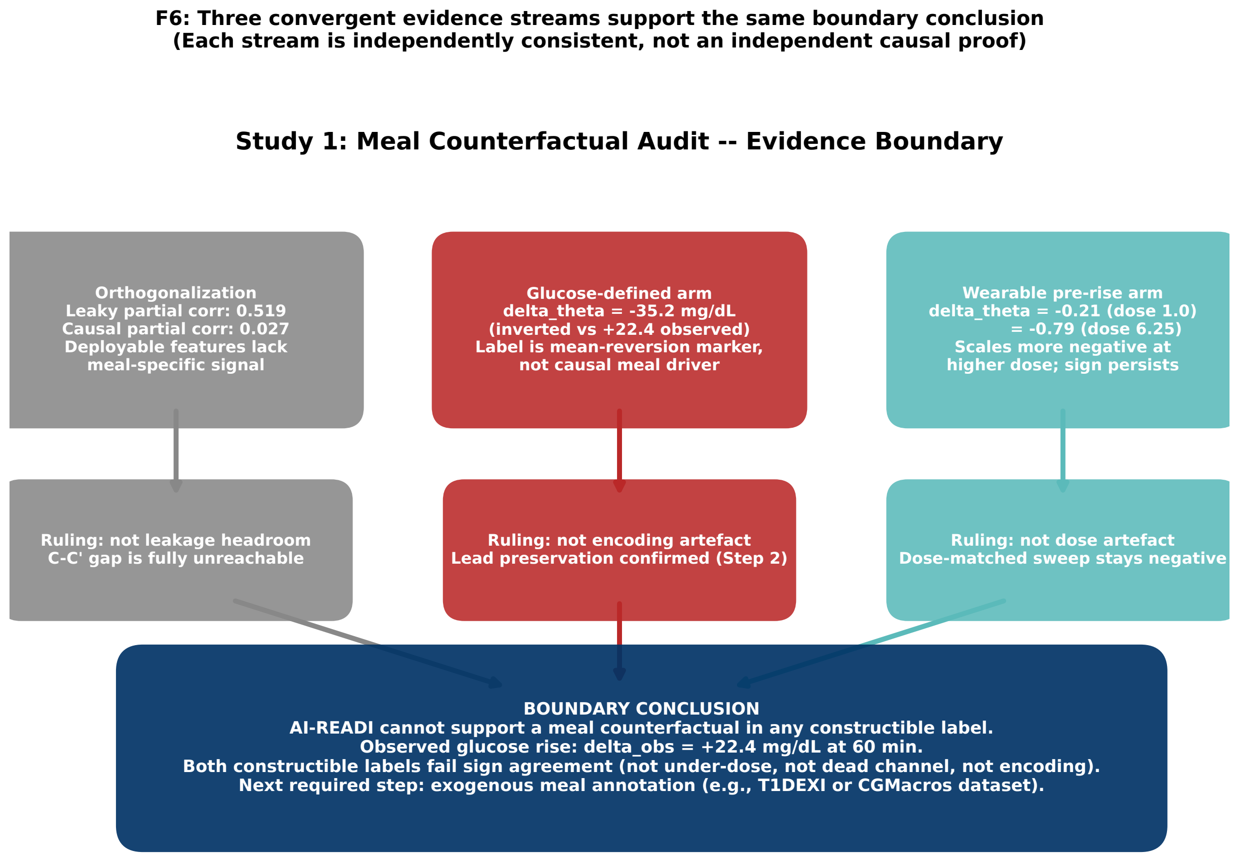

Figure 17: F6: Three convergent evidence streams supporting the boundary conclusion. Each stream rules out one alternative explanation (leakage headroom, encoding artefact, dose artefact), and all three converge on the same verdict. The streams are independently consistent; they are not independent causal proofs. The boundary conclusion follows from their convergence, not from any single test.

Boundary conclusion. AI-READI cannot support a meal counterfactual in any constructible label given the current dataset. The observed glucose rise is $\Delta_{\text{obs}} = +22.2$ mg/dL at 60 min. Both constructible labels fail sign agreement at every dose and every operating point. The failure is not attributable to dead channels (G_{ff} was successfully revived), dose magnitude (dose-matched sweep confirms), or encoding artefacts (lead preservation was confirmed in Step 2). The required next step is exogenous meal annotation from a dataset with prospective logging (for example, T1DEXI or CGMacros), which would enable a label that carries pre-rise causal information not derivable from the CGM slope alone.

10.4 Study Framing

This report presents one of three planned studies.

Study 1: Meal Counterfactual Audit (this section). Concludes that AI-READI does not support a meal counterfactual with any constructible label. The boundary is established by three convergent checks: orthogonalization, glucose-defined arm inversion, and wearable arm dose-scaled null. The recommended next step is evaluation on a dataset with exogenous meal annotations.

Study 2: Insulin and Medication Counterfactuals. Asks whether the scenario-aware channel supports a valid insulin-dose or medication-adherence counterfactual on AI-READI, where external covariates provide labeling that does not depend on the CGM slope. The insulin quarantine and the low-confidence medication flags established in this report define the scope of that evaluation.

Study 3: Downstream Clinical Decision Support. Asks whether the SSM-CGM streaming distribution, with or without a valid counterfactual channel, supports reliable hyperglycemia or hypoglycemia risk alerts at the operating points identified by the pre-rise detector. This study uses the hyperglycemia AUPRC and recall endpoints already reported in Table 23 as its primary metrics.

11 Counterfactual Strategy for AI-READI SSMCGM-Stream

11.1 Motivation and scope

The goal of the counterfactual component is to move SSMCGM-Stream from passive forecasting toward controlled, behavior-aware simulation. However, the AI-READI setting is fundamentally different from T1DEXI [1, 8]. T1DEXI contains explicit action streams such as timed insulin boluses, carbohydrate intake, and insulin-on-board, whereas AI-READI provides CGM, wearable signals, static clinical metadata, and a predicted meal flag. Therefore, AI-READI does not support insulin-dose or carbohydrate-dose counterfactuals. The variable `med_insulin` is treated as a static participant descriptor, not as an editable action. All AI-READI counterfactuals are therefore framed as *proxy scenario simulations*: meal-like, activity-like, stress-like, and sleep/rest physiological states.

The current SSMCGM-Stream run already demonstrates a strong participant-held-out forecaster, but its proxy scenario branch is not yet scientifically interpretable. Forecast-only and factual-future metrics are nearly identical, and the measured mean absolute scenario deltas are close to zero. This motivates a conservative strategy: before interpreting any scenario as a plausible counterfactual effect, we must first prove that the model uses the future scenario channel, that the edited scenarios lie on the empirical AI-READI manifold, and that model-implied effects agree with observed natural experiments.

The proposed counterfactual strategy has five linked components:

1. a scenario-aware streaming forecasting model with explicit forecast-only and factual-future pathways;
2. behavior-level, multivariate scenario edits rather than single-variable perturbations;
3. trajectory-level support checks to enforce positivity and avoid out-of-distribution scenario paths;
4. natural-experiment validation against matched observed windows;
5. sensitivity analysis for unmeasured confounding, with additive bias bounds for continuous glucose effects and E-values for binary clinical risks.

This section describes the mathematical and machine-learning framework behind these components.

11.2 Streaming state-space forecasting backbone

Let participant i have static covariates

$$s_i = \{\text{age}, \text{HbA1c}, \text{BMI}, \text{site}, \text{medications}, \ldots\}. \tag{8}$$

These are encoded by a static encoder g_ϕ into an embedding

$$e_i = g_\phi(s_i). \tag{9}$$

At each 5-minute time step t, the dynamic observation vector $x_{i,t}$ contains observed CGM and wearable variables up to the current time, for example CGM, heart rate, steps, stress, and sleep stage. The streaming SSM update is

$$h_{i,t} = f_\theta(h_{i,t-1}, x_{i,t}, e_i), \tag{10}$$

where $h_{i,t}$ is the participant-specific hidden state after observing the history up to time t. In the current SSMCGM-Stream implementation, f_θ is based on selective state-space/Mamba blocks [4, 5]. The practical advantage of this formulation is that the historical context is compressed into a fixed-dimensional state, so multiple future scenarios can be decoded from the same $h_{i,t}$ without re-encoding the full historical window.

For a forecast horizon of H steps, the decoder receives three groups of future inputs:

- $K_{t+1:t+H}$: known future inputs, such as time-of-day and calendar encodings;
- $A_{t+1:t+H}$: future scenario/proxy variables, such as meal proxy, activity footprint, stress, or sleep/rest signals;
- $M_{t+1:t+H}$: masks indicating whether scenario variables are known, unknown, factual, or edited.

The model predicts glucose quantiles through a residual-current target,

$$\hat{y}^{(q)}_{i,t+h} = y_{i,t} + \hat{r}^{(q)}_{i,t,h}, \qquad h = 1, \ldots, H, \tag{11}$$

where $y_{i,t}$ is the current observed glucose and $\hat{r}^{(q)}_{i,t,h}$ is the predicted residual quantile at horizon h. This makes persistence a strong baseline and prevents cold-start forecasts from predicting absolute glucose from scratch.

11.3 Forecast-only and factual-future pathways

The first requirement for counterfactual modeling is to check whether the scenario channel is alive. We therefore evaluate two decoder pathways for the same hidden state $h_{i,t}$.

The forecast-only pathway hides future scenario information:

$$\hat{Y}^{fo}_{i,t+1:t+H} = D_\theta(h_{i,t}, K_{t+1:t+H}, A^0_{t+1:t+H}, M^0_{t+1:t+H}), \tag{12}$$

where A^0 is set to zero or neutral values and M^0 marks the future scenario path as unknown.

The factual-future pathway reveals observed future wearable/proxy variables retrospectively:

$$\hat{Y}^{ff}_{i,t+1:t+H} = D_\theta(h_{i,t}, K_{t+1:t+H}, A^{obs}_{t+1:t+H}, M^{obs}_{t+1:t+H}). \tag{13}$$

This pathway is not deployable, because it uses information from the future. Its role is diagnostic: if the factual-future pathway does not improve prediction on event-rich windows, then the model has not learned to use scenario information.

Using the pinball loss $\rho_q(u) = \max(qu, (q-1)u)$, the multi-horizon quantile losses are

$$\mathscr{L}_{fo} = \sum_{q\in\mathscr{Q}} \sum_{h=1}^{H} \rho_q \left(y_{i,t+h} - \hat{y}^{fo,(q)}_{i,t+h} \right), \tag{14}$$

$$\mathscr{L}_{ff} = \sum_{q\in\mathscr{Q}} \sum_{h=1}^{H} \rho_q \left(y_{i,t+h} - \hat{y}^{ff,(q)}_{i,t+h} \right). \tag{15}$$

The factual-future gain is

$$G_{ff} = \mathscr{L}_{fo} - \mathscr{L}_{ff}. \tag{16}$$

A useful scenario channel should satisfy $G_{ff} > 0$ on event-rich anchors. If $G_{ff} \approx 0$, any inference-time scenario edit will mostly be arbitrary because the decoder was trained to ignore the future scenario path.

11.4 Scenario-aware training objective

To make the scenario path useful, the training objective should include both forecast-only and factual-future losses:

$$\mathscr{L}_{train} = \mathscr{L}_{fo} + \lambda_{ff}\mathscr{L}_{ff} + \lambda_{gain}\mathscr{L}_{gain} + \lambda_{bal}\mathscr{L}_{bal}. \tag{17}$$

The gain term is applied only to event-rich anchors, where future wearable/proxy information is expected to matter:

$$\mathscr{L}_{gain} = \mathbb{1}_{event} \left[m + \mathscr{L}_{ff} - \mathscr{L}_{fo} \right]_+, \tag{18}$$

where $m \geq 0$ is a small margin. This loss penalizes the model when the factual-future path is not at least m better than the forecast-only path. Importantly, we do *not* include any term of the form $-\mathbb{E}|\Delta|$. Such a term would reward large scenario effects and could manufacture fake counterfactuals. The model is encouraged to use real future information when it is predictive, but it is never directly rewarded for making scenario effects large.

The optional balancing term $\mathscr{L}_{bal}$ is inspired by counterfactual recurrent networks [3]. Let B_t denote an observed behavior-proxy label, such as high activity, sleep/rest, stress, or meal-proxy. A behavior classifier c_ψ attempts to predict B_t from the hidden history representation $h_{i,t}$:

$$\hat{B}_t = c_\psi(h_{i,t}). \tag{19}$$

A domain-adversarial objective can be written as

$$\min_\theta \max_\psi \left[\mathscr{L}_{train} - \lambda_{bal}\, CE(B_t, c_\psi(h_{i,t}))\right]. \tag{20}$$

The purpose is to reduce the association between patient history and behavior assignment, which is a source of time-dependent confounding. This term should be introduced only after simpler two-pathway training and event oversampling have been tested, because excessive balancing can degrade forecasting performance.

11.5 Behavior-level scenarios rather than symptom edits

A central design choice is to define scenarios at the level of behaviors, not individual wearable variables. Directly setting heart rate to a high value is not a valid intervention on exercise: heart rate is a physiological consequence of many possible causes, including activity, stress, illness, caffeine, and glucose dynamics. Therefore, AI-READI scenarios are represented by behavior labels

$$B \in \{\text{meal\_proxy}, \text{mild\_activity}, \text{high\_activity}, \text{sleep\_rest}, \text{sedentary\_stress}\}. \tag{21}$$

Each behavior is realized as a multivariate wearable footprint:

$$B \longrightarrow (\text{steps}, \text{heart rate}, \text{stress}, \text{sleep stage}, \text{awake}, \ldots). \tag{22}$$

This avoids impossible single-variable counterfactuals and is closer to a target-trial emulation view: the intervention is a behavioral strategy, and the wearable variables are its observed proxy signature.

11.6 On-manifold empirical scenario path generation

A first attempt could set scenario variables using independent percentiles, for example steps at the participant's 75th percentile and heart rate at the 65th percentile. This is useful for prototyping, but it can create impossible combinations because marginally plausible values are not necessarily jointly plausible. The preferred approach is empirical conditional path sampling, related in spirit to controllable sequence editing for counterfactual generation [7].

For each anchor (i,t), define the matching context

$$C_{i,t} = [y_{i,t-L:t}, \Delta y_{i,t-L:t}, HR_{i,t-L:t}, steps_{i,t-L:t}, stress_{i,t-L:t}, sleep_{i,t-L:t}, hour(t), s_i]. \tag{23}$$

For a behavior B, the future scenario path is sampled from observed training windows with similar context:

$$A^B_{i,t+1:t+H} \sim p_{train}(A_{u+1:u+H} \mid B_u = B, C_u \approx C_{i,t}). \tag{24}$$

In practice, this can be implemented by nearest-neighbor retrieval, cluster prototypes, or weighted averaging among observed analogue windows. This makes the generated scenario path lie closer to the empirical AI-READI manifold than independently constructed percentiles.

The actual edit is localized in time and scope. For a behavior B starting at horizon step τ and lasting d steps, the scoped edit operator is

$$(A^B, M^B, S^B) = \mathscr{E}_{B,\tau,d}(A, M, S; C_{i,t}), \tag{25}$$

where S is a source/type variable distinguishing unknown future values, factual future values, planned proxy values, and edited proxy scenarios. For affected variables $j \in \mathscr{V}_B$,

$$A^B_{t+h,j} = \begin{cases} \tilde{A}^B_{h,j}, & h \in [\tau, \tau+d],\ j \in \mathscr{V}_B, \\ A_{t+h,j}, & \text{otherwise}, \end{cases} \tag{26}$$

where $\tilde{A}^B$ is the empirical analogue path. The mask is set to one for edited variables during the intervention interval, and the source/type label marks them as edited proxy values.

11.7 Scenario predictions and model-implied effects

Given the same hidden state $h_{i,t}$, the forecast-only prediction is

$$\hat{Y}^{fo}_{i,t+1:t+H} = D_\theta(h_{i,t}, K, A^0, M^0, S^0), \tag{27}$$

whereas the behavior-scenario prediction is

$$\hat{Y}^{B}_{i,t+1:t+H} = D_\theta(h_{i,t}, K, A^B, M^B, S^B). \tag{28}$$

The model-implied proxy scenario effect at horizon h is

$$\Delta_\theta(B,i,t,h) = \hat{y}^{B}_{i,t+h} - \hat{y}^{fo}_{i,t+h}. \tag{29}$$

This is not automatically a causal effect. It is the change in the model's prediction when the future proxy path is changed while holding the observed history and current state fixed. It becomes scientifically interpretable only if the scenario channel is used, the scenario has empirical support, and the model-implied effect agrees with observed natural experiments.

A useful decomposition is

$$\hat{y}^{B}_{i,t+h} = \hat{y}^{base}_{i,t+h} + \Delta_\theta(B,i,t,h), \tag{30}$$

where $\hat{y}^{base}$ captures the forecast from history and known future time variables, and Δ_θ captures the additional scenario-dependent component. This decomposition should be used diagnostically: if $\mathbb{E}|\Delta_\theta| \approx 0$, the scenario branch is inactive or weakly used.

11.8 Trajectory-level support and positivity

Longitudinal counterfactual identification requires positivity: each scenario should have nonzero probability among participants with comparable histories. In this application, this means the edited future wearable path must resemble observed AI-READI paths. Support should therefore be assessed over the whole trajectory rather than at a single time point.

Let $\Psi(A_{t+1:t+H}, C_t)$ summarize a future path and its context, including total steps, maximum heart rate, mean stress, sleep fraction, time of onset, glucose level, glucose slope, time of day, site, HbA1c quartile, and study group. Define the path support distance

$$d_{path}(B,i,t) = \min_{u \in \mathscr{D}^B_{train}} \left\| \Psi(A^B_{i,t+1:t+H}, C_{i,t}) - \Psi(A^{obs}_{u+1:u+H}, C_u) \right\|_{\Sigma^{-1}}. \tag{31}$$

The support class is then

$$support(B,i,t) = \begin{cases} \text{supported}, & d_{path} < c_1 \text{ and } n_{neighbors} > n_{min}, \\ \text{weak}, & d_{path} < c_2, \\ \text{unsupported}, & \text{otherwise}. \end{cases} \tag{32}$$

Only supported scenarios are used in the main counterfactual analysis. Weakly supported scenarios are reported in the appendix. Unsupported scenarios are excluded from effect interpretation.

11.9 Natural-experiment validation

Because AI-READI is observational, model-implied effects must be compared against real observed windows that resemble the proposed scenario. For each behavior B, define a treatment indicator

$$T^B_{i,t} = 1 \tag{33}$$

when the observed future window contains the behavior footprint, and $T^B_{i,t} = 0$ for matched control windows. For example, an activity treated window may require high steps, elevated heart rate, and awake state; a control window should have low steps or normal heart rate while matching on pre-onset glucose and clinical context.

Matching should use rich pre-onset histories, not only current glucose and one slope. The matching vector is

$$H_{i,t}^{match} = [y_{i,t-L:t}, \Delta y_{i,t-L:t}, HR_{i,t-L:t}, steps_{i,t-L:t}, stress_{i,t-L:t}, sleep_{i,t-L:t}, hour(t), s_i]. \tag{34}$$

The observed natural-experiment effect is

$$\Delta_{obs}(B,h) = \mathbb{E}\left[y_{i,t+h} - y_{i,t} \mid T_{i,t}^{B} = 1,\ matched\right] \tag{35}$$

$$- \mathbb{E}\left[y_{i,t+h} - y_{i,t} \mid T_{i,t}^{B} = 0,\ matched\right]. \tag{36}$$

The model-implied effect is averaged over the same supported anchors:

$$\Delta_{\theta}(B,h) = \mathbb{E}_{(i,t)\in\mathscr{S}_B}\left[\hat{y}_{i,t+h}^{B} - \hat{y}_{i,t+h}^{fo}\right], \tag{37}$$

where $\mathscr{S}_B$ is the set of supported anchors for behavior B.

Validation metrics include effect MAE,

$$EffectMAE(B) = \frac{1}{H}\sum_{h=1}^{H} |\Delta_{\theta}(B,h) - \Delta_{obs}(B,h)|, \tag{38}$$

and sign agreement,

$$SignAgreement(B) = \frac{1}{H}\sum_{h=1}^{H} \mathbb{1}[\operatorname{sign}(\Delta_{\theta}(B,h)) = \operatorname{sign}(\Delta_{obs}(B,h))]. \tag{39}$$

The primary validation report should include treated-window counts, control-window counts, support coverage, sign agreement, and effect MAE. Calibration slope is useful as a secondary metric, but it should not replace interpretable diagnostics.

11.10 Sensitivity analysis for unmeasured confounding

Sequential ignorability cannot be verified in AI-READI. Behavior may be confounded by unobserved intent, symptoms, meals, insulin, illness, or other latent factors. Sensitivity analysis is therefore a core component of the counterfactual framework rather than an optional add-on [6].

For continuous glucose effects in mg/dL, we use an additive bias bound. Let δ denote a hypothetical unmeasured confounding bias. The adjusted observed effect is

$$\Delta_{obs}^{adj}(B,h;\delta) = \Delta_{obs}(B,h) - \delta. \tag{40}$$

We evaluate a grid such as

$$\delta \in \{-20,-15,-10,-5,0,5,10,15,20\}\ \text{mg/dL}. \tag{41}$$

The sign-flip threshold is

$$\delta_{flip}(B,h) = \min_{\delta}\left\{\operatorname{sign}(\Delta_{obs}^{adj}(B,h;\delta)) \neq \operatorname{sign}(\Delta_{obs}(B,h))\right\}. \tag{42}$$

This gives a direct interpretation: how large an unmeasured glucose-scale bias would need to be to reverse the estimated effect.

For binary clinical outcomes, such as hypoglycemia within H minutes,

$$Y_{i,t,H}^{hypo} = \mathbb{1}\left[\min_{1\le h\le H} y_{i,t+h} < 70\right], \tag{43}$$

or hyperglycemia,

$$Y_{i,t,H}^{hyper} = \mathbb{1}\left[\max_{1\le h\le H} y_{i,t+h} > 180\right], \tag{44}$$

we estimate risk ratios comparing treated and matched controls and report E-values [9]. E-values are appropriate for risk-ratio-scale sensitivity analysis; they should not be used as the main sensitivity tool for continuous mg/dL effects unless those effects are converted into binary clinical risks.

11.11 Attribution and channel-use diagnostics

Attribution is useful only after the behavioral tests show that the scenario channel matters. Raw gradients such as

$$\left\| \frac{\partial \hat{y}_{t+h}}{\partial A_{t+1:t+H}} \right\| \tag{45}$$

can be unstable in gated SSMs. Therefore, the primary channel-use diagnostic remains the factual-future gain G_{ff} on event windows. Attribution should be used as a secondary explanation tool.

When applied, attribution should answer three questions:

1. Does the decoder place non-negligible attribution on the scenario path rather than only on current glucose?
2. Does attribution increase after the planned scenario onset τ?
3. Does attribution vary by scenario and horizon in a physiologically plausible way?

The hidden-attention interpretation of Mamba models can support this analysis by converting the selective SSM recurrence into an attention-like view of which past or scenario tokens influence the forecast [2].

11.12 Counterfactual horizon design

The current forecasting headline uses a 60-minute horizon, but some behavior effects may unfold over longer periods. Activity, stress, and sleep/rest effects can be delayed or sustained, and meal-like effects may peak beyond the first hour depending on timing. Therefore, the counterfactual validation should be run at several horizons:

$$H \in \{12, 24, 48, 72\} \tag{46}$$

corresponding to 1, 2, 4, and 6 hours at 5-minute resolution. The 1-hour horizon remains the deployment-oriented forecast benchmark, while 2–6 hour horizons test whether behavioral scenario effects are hidden by the short forecasting window.

11.13 Experimental roadmap

The proposed implementation is divided into four phases.

Phase 0: audit the current model without retraining. Compute factual-future gain G_{ff} and scenario deltas by all anchors and by event-rich anchors: meal-proxy windows, high-activity windows, sleep transitions, stress windows, and large glucose-change windows. If $G_{ff} \approx 0$, the current scenario decoder should not be interpreted.

Phase 1: train a scenario-aware stream model. Train variants with the two-pathway objective in Equation (17), event-enriched sampling, and optional source/type embeddings. The primary metric is not large scenario delta; it is positive factual-future gain on event windows.

Phase 2: replace hand-set scenarios with empirical on-manifold scenario paths. Compare independent percentile scenarios, nearest-neighbor empirical paths, and cluster-prototype paths. Report support coverage, path distance, effect stability, and scenario effect by horizon.

Phase 3: validate against natural experiments. For each behavior, compare model-implied effects with observed matched effects using sign agreement and effect MAE. Add continuous sensitivity bounds for mg/dL effects and E-values for binary hypo/hyper risk outcomes.

Phase 4: optional CRN-style balancing. If natural-experiment validation suggests substantial time-dependent confounding, add adversarial behavior balancing. This should be treated as an ablation because too much balancing may reduce forecast accuracy.

Table 25: Matched natural-experiment effect $\Delta_{obs}(h)$ for the `wearable_plus_shape` detector on the non-insulin cohort. Bootstrap 95% CI from participant resampling ($n_{\text{boot}} = 300$). Δ_θ is not available because the scenario channel is dead.

Horizon h	Δ_{obs} (mg/dL)	95% CI	n pairs
60 min	+22.2	[21.6, 22.8]	7,480
120 min	+24.2	[23.5, 25.0]	7,432
240 min	+3.6	[2.8, 4.5]	7,355
360 min	+0.3	[−0.5, 1.0]	7,302

Table 26: Sensitivity analysis for the `wearable_plus_shape` natural experiment. Sign-flip threshold is the smallest additive bias that reverses the sign of Δ_{obs}. E-values use the VanderWeele-Ding formula [9] for the binary hyperglycemia risk ratio; they are reported only for the binary endpoint, not for the continuous glucose effect.

Horizon	Δ_{obs}	Sign-flip δ_{flip}	E-value (hyper)	RR (hyper)
60 min	+22.2 mg/dL	22.2 mg/dL	3.80	2.19
120 min	+24.2 mg/dL	24.2 mg/dL	3.33	1.96
240 min	+3.6 mg/dL	3.6 mg/dL	1.84	1.26
360 min	+0.3 mg/dL	0.3 mg/dL	—	—

11.14 Phase 0 Results: Wearable Counterfactual Readiness Audit

We ran the five-block readiness evaluation for the `wearable_plus_shape` detector on the non-insulin cohort (9,667 unique meal onsets, 1,591 participants). The purpose is not to claim causal validity, but to document precisely where the current system stands relative to the four-phase roadmap.

Block A: channel gate. The factual-future gain is $G_{ff} = 1.8 \times 10^{-5}$ overall, and the `meal_proxy_event` event-window evaluation has zero anchors (`predmeal_flag` future mask coverage is zero). The scenario channel is therefore *dead*: the decoder was trained without a factual-future loss term on meal windows, so any inference-time wearable scenario edit produces near-zero prediction deltas regardless of detector quality. Scenario-aware retraining (Phase 1) is required before the channel can be used.

Block B: matched natural experiment. Despite the dead channel, the matched natural experiment recovers a clear physiological signal. We matched 7,825 treated episodes (`wearable_plus_shape` detected, non-insulin) against 7,538 controls from the same participant in the same two-hour clock bin, with pre-onset glucose within 25 mg/dL and no nearby meal. The observed effect $\Delta_{obs}(h)$ is shown in Table 25.

The effect profile follows classic postprandial glucose kinetics: rapid rise peaking at 120 minutes, returning to baseline by 360 minutes. At 360 minutes the confidence interval crosses zero, consistent with full resolution. The detector is capturing real meal events, not noise.

Block C: trajectory support. Of 7,825 treated anchors, 7,814 (99.9%) are classified as supported; 11 are unsupported due to missing CGM context. The positivity condition is essentially satisfied: the detector fires in states well-represented in training data, so future counterfactual comparisons will not extrapolate into uncharted CGM territory.

Block D: sensitivity to unmeasured confounding. Table 26 reports the additive-bias sign-flip threshold $\delta_{flip}(h) = |\Delta_{obs}(h)|$ and the VanderWeele-Ding E-value for the binary hyperglycemia endpoint.

At 60 and 120 minutes the effect is robust: an unmeasured confounder would need to shift glucose by more than 22 mg/dL (or be associated with both wearable state and hyperglycemia risk by a factor of 3.3–3.8) to explain away the finding. At 240 minutes the sign-flip threshold drops to 3.6 mg/dL, making the effect fragile to modest unmeasured confounding. At 360 minutes Δ_{obs} is +0.3 mg/dL with a CI crossing zero; no E-value is reported because the continuous effect is indistinguishable from zero.

Block E: detection quality linkage. The `wearable_plus_shape` detector provides the most and best-timed treated episodes (0-minute median latency, 46.8% pre-rise) compared to all other setups and CGM-only baselines (35–40 minute post-hoc detection). The higher false-positive rate (11.5 FP/day) dilutes the natural-experiment effect toward zero, making the $+22$ mg/dL finding conservative rather than inflated.

Phase 0 conclusion. The wearable detector captures real pre-meal glucose events with near-perfect trajectory support and robust sensitivity at the 1–2 hour horizons. It is ready to serve as the treated indicator in a natural experiment and as the input to a scenario-aware decoder once the channel is live. The single limiting factor is the dead scenario channel. Phase 1 (scenario-aware retraining) is the required next step; improving the detector further is not.

11.15 Expected outputs

The counterfactual pipeline should produce the following report-ready outputs:

- `scenario_channel_audit.csv`: factual-future gain, event-window gain, and scenario delta by mode;
- `scenario_support.csv`: support class, path distance, and neighbor counts by scenario;
- `scenario_effects_by_horizon.csv`: $\Delta_\theta(B,h)$ by scenario and horizon;
- `natural_experiment_effects.csv`: treated/control counts, Δ_{obs}, Δ_θ, sign agreement, and effect MAE;
- `sensitivity_bounds.csv`: additive bias bounds for continuous effects and E-values for binary clinical risks;
- figures for factual-future gain, support coverage, model-vs-observed effects, and sensitivity thresholds.

11.16 Interpretation rules

The final interpretation should follow strict rules:

1. If $G_{ff} \leq 0$ on event windows, the scenario channel is considered inactive; do not interpret scenario deltas.
2. If a scenario is unsupported by the trajectory-level support check, exclude it from the main counterfactual results.
3. If model-implied effects disagree with observed natural-experiment effects, report the scenario as a failed or unvalidated proxy.
4. If continuous effects are sensitive to small additive bias, state that the result is fragile to unmeasured confounding.
5. E-values are reported only for binary clinical risk endpoints, not as the main sensitivity metric for continuous glucose effects.
6. AI-READI counterfactuals should never be described as insulin-dose or carbohydrate-dose counterfactuals.

Under these rules, the contribution is not a claim that AI-READI already supports definitive causal recommendations. The contribution is a validated framework for deciding which proxy behavioral scenarios are meaningful, which are unsupported, and which require additional action labels or model supervision.

12 Clinical Safety

Median hypoglycemia alarms are conservative: precision is high but recall is low. The q0.1 risk rule substantially increases hypoglycemia recall at the cost of lower precision, which is the expected clinical safety trade-off. Hyperglycemia is more predictable than hypoglycemia, but the q0.9 risk rule similarly trades precision for sensitivity.

Tail errors remain clinically important despite good average MAE. The forecast-only p90, p95, and p99 absolute errors are 24.25, 34.05, and 59.69 mg/dL, respectively. True TIR is 89.7% and predicted TIR is 91.6%, giving a 1.86 percentage-point overprediction of time in range.

Table 27: Hypoglycemia and hyperglycemia event detection using median and risk-quantile alarm rules.

Event	Horizon	Rule	Prec.	Recall	Spec.	F1
Hypo < 70	overall	median_q50_lt_70	0.734	0.231	1.000	0.352
Hypo < 70	30 min	median_q50_lt_70	0.651	0.147	1.000	0.239
Hypo < 70	60 min	median_q50_lt_70	0.610	0.038	1.000	0.072
Hypo < 70	overall	risk_q10_lt_70	0.314	0.585	0.995	0.409
Hypo < 70	30 min	risk_q10_lt_70	0.291	0.571	0.994	0.385
Hypo < 70	60 min	risk_q10_lt_70	0.224	0.349	0.995	0.273
Hyper > 180	overall	median_q50_gt_180	0.886	0.781	0.988	0.830
Hyper > 180	30 min	median_q50_gt_180	0.879	0.777	0.987	0.825
Hyper > 180	60 min	median_q50_gt_180	0.833	0.642	0.984	0.726
Hyper > 180	overall	risk_q90_gt_180	0.610	0.935	0.926	0.738
Hyper > 180	30 min	risk_q90_gt_180	0.614	0.935	0.927	0.741
Hyper > 180	60 min	risk_q90_gt_180	0.484	0.887	0.883	0.626

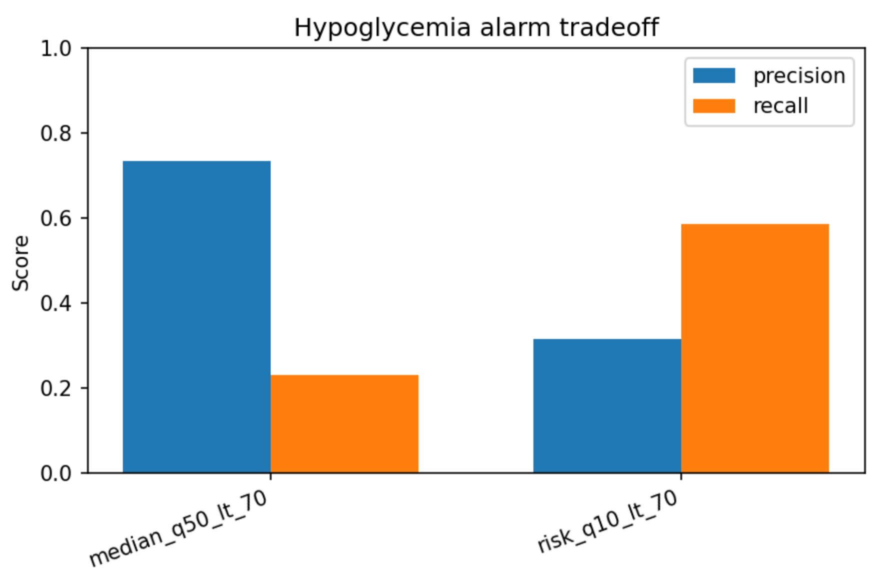

Figure 18: Clinical alarm trade-off. Median forecasts are conservative for hypoglycemia, while the q0.1 risk rule increases recall at the cost of lower precision.

13 Interpretability

The most defensible interpretability results for this AI-READI run are the subgroup, persistence, and scenario-delta diagnostics. The residual-current target means a model can look strong by staying close to persistence, so the matched persistence baseline is essential for interpretation. The model does beat persistence, but the margin is modest, which is expected for 60-minute CGM forecasting.

The proxy scenario branch is not currently interpretable as a physiological what-if mechanism: prediction deltas are near zero, and `predmeal_flag` is unavailable in the audited future horizon for the test anchors. No insulin-on-board equivalent is available in AI-READI, and no insulin causal interpretation should be included.

14 Ablation and Runtime

The stateful Mamba run is feasible on a single GPU. The best validation checkpoint occurs at epoch 5; later epochs improve training loss but do not improve validation pinball, so the best checkpoint rather than the final epoch is used for evaluation. The ablation launcher prepares short probes, but full ablations remain a follow-up experiment.

Table 28: Runtime, memory, and training-throughput summary from the selected AI-READI stream run.

Quantity	Value
Best validation epoch	5
Best validation pinball	3.286 mg/dL
Max training throughput	4868 anchors/s
Peak training GPU memory	897 MB
Evaluation GPU memory	307 MB
Evaluation CPU RSS	8699 MB
Median update latency	3.76 ms
Mean 1h forecast latency	0.0037 ms

Table 29: Ablation status. Some ablations are configured as probes but have not yet been fully run.

Ablation	Status	Interpretation
Full residual-current model	configured	available as reference
Absolute-target / no residual-current	prepared	not run by current pipeline
No static h0/FiLM	probe supported	run with launch_ablation_probes.sh
No scenario masks	prepared	current model lacks safe switch
No scenario decomposition	probe supported	run with launch_ablation_probes.sh

[TODO: No runtime backend figure was found; runtime is summarized in Table 28.]

The current ablation table is a status table, not a result table. The full model, no-static-FiLM, and no-decomposition probes can be launched with the provided script. Absolute-target and no-scenario-mask variants require additional implementation before their numbers can be interpreted.

15 Discussion and Limitations

The AI-READI port produces a viable participant-held-out streaming forecaster and improves over persistence on matched anchors. The result is scientifically meaningful because the split is participant-held-out and the recurrent state never crosses segment boundaries. The improvement over persistence is real but not huge, which is an important limitation of residual-current CGM forecasting: current glucose already carries most of the short-horizon signal.

The strongest current claims are forecasting performance, persistence improvement, participant-level subgroup behavior, and q0.1/q0.9 safety-alarm trade-offs. The weakest current claims are personalization and proxy scenarios.

Matched-anchor personalization is flat in the current artifacts, and offset correction worsens MAE. Proxy scenario deltas are near zero, so the scenario branch is not yet scientifically interpretable.

Limitations are specific to AI-READI. The dataset lacks timed insulin dose, insulin-on-board, and carbohydrate quantity logs; `med_insulin` is static metadata only. The predicted meal flag is noisy and may be retrospective. The residual-current target can hide weak dynamic learning if persistence is not reported. Participant-level averages are necessary because anchor-weighted metrics overrepresent participants with longer clean segments. Finally, site and phenotype subgroup effects require follow-up audits before being interpreted as biological effects rather than cohort composition or domain shift.

A Compute Placement: GPU vs. CPU

CUDA accelerates training through the Mamba/Triton scan backend. The streaming recurrence and decoder are still compatible with CPU execution for portability, but the reported full-cohort training runs used one NVIDIA A100 GPU.

B Generated Appendix Tables and Figures

Table 30: Full training history for the 10-epoch resumed stateful Mamba run.

Epoch	Train	Val	Anchors/s	Seconds
0	3.563	3.423	3089	241.8
1	3.394	3.330	4714	158.4
2	3.339	3.298	4709	158.6
3	3.304	3.325	4693	159.1
4	3.280	3.316	4720	158.2
5	3.263	3.286	3241	230.4
6	3.243	3.321	4829	154.7
7	3.225	3.306	4831	154.6
8	3.213	3.300	4868	153.4
9	3.197	3.319	4843	154.2

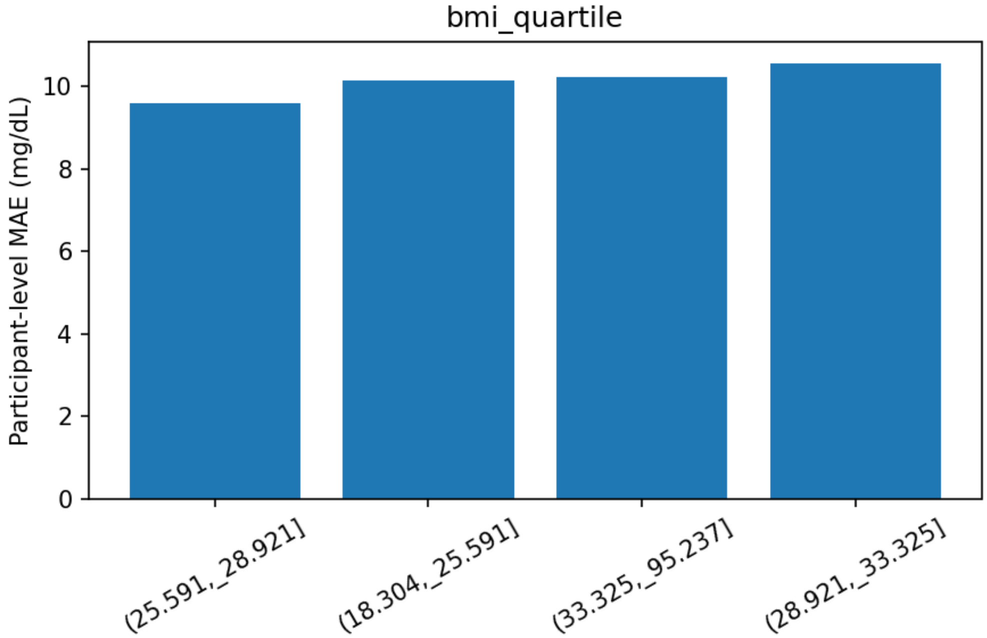

Figure 19: Appendix subgroup plot: participant-level MAE by BMI quartile.

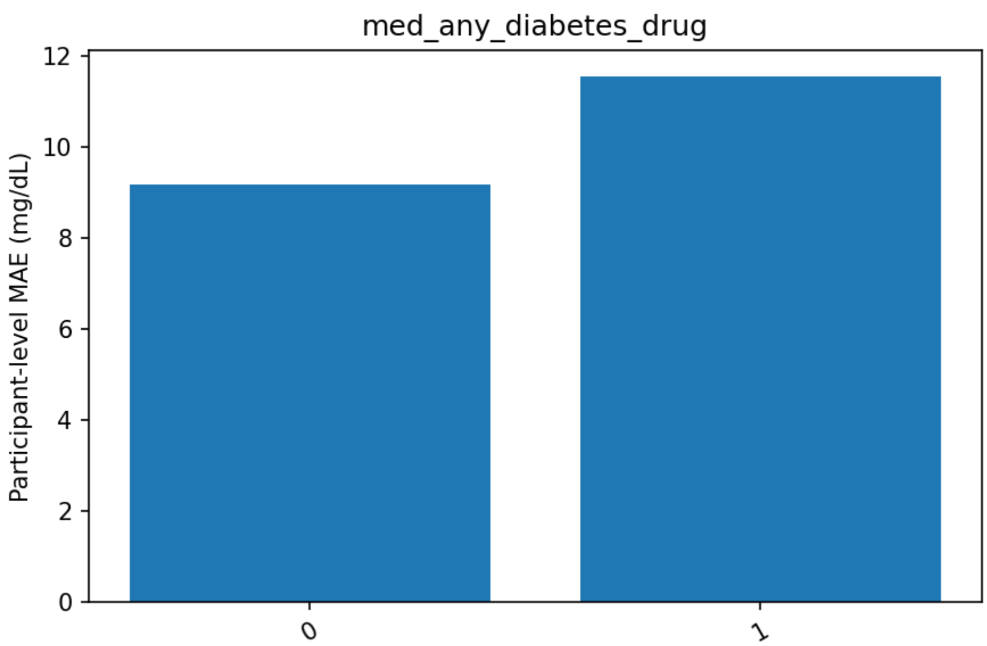

Figure 20: Appendix subgroup plot: participant-level MAE by any diabetes-drug metadata flag.

C Generated Results Snapshot

Scalar result macros are loaded from `sections/generated_results_summary.tex`. The file is retained with the report source for provenance, while the main appendix keeps the generated tables focused on model results rather than macro bookkeeping.

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